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Travellers, trainees, and urban transformations: knowledge circulation in the pre-industrial Stockholm metal trades, c. 1750–1820

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ABSTRACT

Urban contexts and town-based groups have long been associated with creativity, innovativeness, and the dissemination of technical know-how. Studies dealing with early modern European towns have recently added important findings on this subject, examining practices of work and knowledge circulation that contributed to improvements in the spheres of craftwork and manufacturing well before the onset of industrialisation. This article builds on such research contributions, and it does so by exploring developments at the fringes of a pre-industrial European economy, namely in Stockholm, Sweden, during the period c. 1750–1820. Using a large collection of sources from the Swedish state archives, and zooming in on the important metalworking sector, it illuminates transnational itineraries as well as localised activities that made the capital into a key node in the Swedish – and indeed European – metal trades. Integrating a large number of working individuals and knowledge-making practices into the discussion, moreover, it provides a good basis for nuancing traditional assumptions of a long stagnation in the Swedish capital during the end of the early modern era.

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Introduction

Reporting to the Swedish Diet in April 1769, Samuel Schröder stressed the worsening state of metal manufacturing in the realm. Sixteen years earlier, he had taken office in the Bureau of Trade (*Kommerskollegium*) as state supervisor for fine metalworking during an expansive period. Manufactories and workshops were founded with the support of special privileges, aiming to establish clear boundaries with existing forms of production such as town-based guild crafts, and financial backing was made available through public loans that, in turn, spurred a rising inflation. This positive trend came to a halt after the Seven Years' War. Schröder described how the Swedish economy was drawn into the trade crisis of 1763 and the political situation became ever more turbulent; a growing critique of the agenda pursued by the ruling Hat Party (*Hattarna*)

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led to the withdrawal of subsidies as well as to restrictions of both 'public and private credit'. After the Diet of 1765–66, the failed attempts by the new political leadership to strengthen the Swedish currency, which in effect hastily turned the escalating inflation into a problematic deflation, had an additional negative impact on the manufacturing trades; the inventories of unsold goods grew, and workers were sacked or left their employments voluntarily. Both rural and urban workplaces succumbed to this 'conjuncture', but the downturn was perhaps most evident in Stockholm. Schröder noted that a dozen workshops in the capital's metal trades had closed between 1764 and 1769, and several manufacturers had left for competing nations. Stockholm-based watchmaking was brought up as an example of a trade facing a quite 'miserable' situation if forceful state measures were not taken.[1]

This negative review must be placed in a larger context. Swedish scholars have debated whether the economic crisis of the 1760s signified the end for state initiatives in the manufacturing sector, or if it rather resulted in a move towards standardisation and more efficient enterprises.[2] Partly abandoning such one-sided explanations, recent work instead stresses the complexity of early modern manufacturing, highlighting the existence of various organisational solutions and credit arrangements.[3] Still, most studies do agree on one point. The Stockholm economy saw a long stagnation from the mid-eighteenth century, linked to a criticism of old monopolies and growing competition from provincial towns, with the capital's declining dominance being particularly evident in the large textile sector. Not until the mid-1800s did Stockholm witness a new wave of industrial expansion, this time in 'capital- and skill-intensive sectors'.[4]

In this regard, examples from the metal trades make for a different perspective, with people's movements and the dissemination of knowledge being brought to the forefront of analysis. As a starting point, we can once again consult Schröder's writings. Some six months after his gloomy report, he witnessed 'the first casting' at a new iron foundry in Bergsund, in the southern parts of the capital, directed by Thomas Lewis. The Scotsman had also led the work of erecting the foundry's air furnace with its 'requisite equipment', all according to 'the way it is usually done in England'. Thus, in Schröder's view, he had shown sufficient 'knowledge and skills in the art of casting' to be worthy of 'some rewards or support'.[5] Lewis later added that he had arrived in Stockholm, after leading a similar project in Gothenburg, with a desire to 'make my knowledge useful in another place'.[6] He was, however, not the only mobile manufacturer in the capital to be recognised by Schröder, and his foundry was one among several promising enterprises at the end of the turbulent 1760s.

Such observations provide the basis for a nuanced discussion of urban manufacturing in the northern periphery of eighteenth-century Europe. This article studies the impact of mobility and the circulation of skills and technical know-how in the Stockholm metal trades before, during, and after the crisis of the 1760s, and it does so by highlighting transnational itineraries and domestic movements as well as local practices of work organisation, training, and testing. This objective resonates well with new micro-spatial perspectives on pre-industrial labour, as well as with recent studies stressing the intricate character of artisanal production and the codification of techniques in early modern societies.[7] Zooming in on knowledge-making activities in one urban area, while at the same time linking them to a larger economic context of change, I show how actors and

workplaces in the 'stagnating metropolis' occupied key roles in nurturing innovation in the Swedish manufacturing sector.[8]

My argument is elaborated in three sections. To begin with, I present more general trends in the context studied, while also linking them to current perspectives in European historical research that form the analytical framework. This discussion is followed by two empirical sections, the first of which deals with the impact of European study tours and cross-border technology transfer before the crisis years. It likewise reveals how these transnational movements were related to training and skill formation in the Swedish capital as well as to regional patterns of workforce mobility. The second empirical section examines the increasingly turbulent situation of the 1760s, when many manufacturers left Stockholm for other nations or provincial towns. I will also scrutinise developments after the harsher 'conjuncture', by targeting the rise of new urban enterprises and the achievements of various mobile practitioners. Both these sections build on a large collection of manuscript sources: protocols, letters, and reports from state archives – like ones related to the Swedish Diet and *Kommerskollegium* – are used alongside information from Samuel Schröder's detailed diaries. Together, these texts make it possible to explore important processes both in-depth and over time, while highlighting the interaction among different groups in the metal trades, thus providing an empirical basis for nuancing the idea of a long urban stagnation.

'Like the heart in a human body'

Eighteenth-century Stockholm was the unrivalled economic centre of the Swedish realm. It was a hub in the expanding iron trade – the only domestic sector that was fully integrated into the European economy – linking provincial iron works (or *bruk*) with customers in Britain and Continental Europe. After the defeat in the Great Northern War, the capital also became a focal point for ambitions to revitalise Swedish industry, through state-supported ventures that often interfered with the established interests of craft guilds and urban traders.[9] The 'system' taking shape during the following decades, to use Schröder's term, shared many features with efforts in other parts of Europe, notably so with French manufacturing policies.[10] 'Free masterships' were introduced at the turn of the 1720s, and, shortly afterwards, the burgeoning manufacturing sector was equipped with jurisdictional bodies of its own, the hallmark courts (*Hallrätter*). Other measures followed during the 1730s, launched in line with decrees of foreign trade that promoted import substitution. Updated general privileges were issued for the manufacturing trades in 1739, along with the creation of a state-run Office of Manufacturing (*Manufakturkontoret*) to handle the distribution of loans, advance payments, and a range of rewards (such as for exports, relocations, and inventions) from a special fund. Meanwhile, the regulation of production, training, and mobility became more detailed, and the administrative surveillance (shared between *Manufakturkontoret*, *Kommerskollegium*, and the local *Hallrätter*) grew in scope.[11]

This agenda mainly favoured the textile trades (the making of woollen and worsted, silk wares, cottons, and linens), which accounted for the largest share of state-supported manufacturing, but it also impacted other sectors such as metalworking.[12] The role of Stockholm in this development was obvious to Swedish commentators. In 1741, Lars Salvius emphasised the increased urban 'life and movement', compared to the difficult

years before the Peace in Nystad, and hailed that the capital was truly 'like the Heart in a Human Body, absorbing and spreading the Blood to all Arteries and Limbs'.[13] Economy professor Anders Berch agreed, stressing that the concentration of industry and commerce, and hence also knowledge and capital, in large towns like Stockholm was one key objective when seeking to encourage a prosperous and well-ordered 'common householding' (*allmän hushållning*) in the Swedish realm.[14]

There was substance in these accounts. Stockholm's guild crafts exhibited a quick recovery after the war, followed by an expansion for the manufacturing trades.[15] Only some ten metal manufacturing enterprises were listed in the *Hallrätt*'s report from 1740, but this number grew steadily during the following decades. In 1760, some 50 workplaces were listed under 'Iron, Steel, and Metal', ranging from small craft shops to a few larger metal works (or *fabriques*), and they were supplemented by close to 20 (more or less) metal-consuming businesses that were instead listed under the headline 'miscellaneous'.[16] The *Hallrätt*-reports are indeed good starting points, although far from complete, and adding information from the head tax records (*mantalslängder*) gives a better picture.[17] All workplaces listed in Table 1 operated (at least formally) outside the guilds' jurisdiction, with manufacturers who produced new products (like trinkets) or imitations of previously imported goods (like fine tools and instruments), worked with novel material combinations and finishing techniques (as in the making of casted and alloyed wares), or used new organisational solutions and work methods (most clearly exemplified by the watch- and clockmaking trade).

To this observation must be added that the capital's guild craftsmen, who remained organised in their respective traditional associations until the liberal reforms of the mid-nineteenth century (in 1846 and 1864), made an equally wide assortment of goods from gold, silver, steel, iron, tin, copper, and brass. Moreover, the urban production landscape incorporated metalworking activities that were associated neither to guilds nor to the manufacturing trades, as clearly demonstrated by the production at the large Royal Mint (*Kungliga Myntet*) and the artillery. The core of the metal trades was the town centre (Nicolai parish), housing several large workplaces such as goldsmiths' workshops and watchmaking *fabriques*, but many workshops were also located in Klara parish, north of the town centre. On a more general level, it can be noted that each of the capital's districts was characterised by an assortment of large and small metalworking businesses as well as by a blend of manufacturing enterprises and guild workshops.

Table 1. Metal manufacturing in Stockholm, 1760. Sources: Stockholms stadsarkiv, Överståthållarämbetet för uppbördsärenden, Kamrerareexpeditionen, G1BA:21, head tax records (*mantalslängder*) Stockholm 1760; Stockholms stadsarkiv, Hall- och manufakturrätten, B3:5, Stockholm *Hallrätt*, report, 1760.

Craft (sorted by main orientation)	Workplaces	Employees
Watchmaking and related gear	14	77
Instruments and tools	19	65
Trinkets, ornaments, jewelry, cases	18	38
Foundry work, alloying, tinning, gilding	13	37
Pins and gear for the textile sector	12	16
Other	16	23
Total	***92***	***256***

Note: 'Employees' includes all workers listed in the records, skilled employees and trainees as well as unskilled workers such as servants and day labourers.

In terms of industrial development, however, it is the subsequent crisis and the notions of stagnation that have made an imprint in economic–historical studies. Johan Söderberg *et al* stress that the period 1760–1850 was marked by a 'de-industrialization process' in the capital, with decreases in output and workforce, declining investments, and a sustained 'informalization of the urban economy'. These trends unfolded in a 'wasteful' demographic system with high mortality rates.[18] The downturn is particularly evident considering the number of workers in the textile sector. The woollen, silk, and cotton branches jointly employed on average 7810 workers per year in the 1750s (86,5 percent of the capital's manufacturing workforce), a number that dropped to 5497 in the 1760s, 4742 in the 1770s, and then continuously declined during the following decades. In the 1840s, the three branches employed 1506 workers annually (41.2 percent of total).[19]

This picture is partly nuanced by the fact that the number of enterprises increased during the late-eighteenth century and the early 1800s, a period that also saw a rising productivity in the textile sector. Moreover, it must be noted that the average annual number of workers outside the three dominant textile branches increased from 1222 in the 1750s (13.5 percent of the total) to 2153 in the 1840s (58.8 percent of the total). Nonetheless, Söderberg *et al* concludes that the manufacturing trades in Stockholm seem to have been characterised by 'a lack of dynamics'.[20] Further confirming such findings, other scholars have noted concurrent tendencies of stagnation in the Stockholm guilds, especially when compared to the expansion in rural areas and provincial towns.[21] Industrial 'life and movement', using Salvius' words, has rather been viewed as rooted in craft activities that were more closely tied to the expanding agrarian economy. Metalworking is one illustrative example. Founded in 1771, the 'free town' (*fristad*) in Eskilstuna became a centre for manufacturing, while rural producers in several regions advanced their positions regarding the making and circulation of iron wares.[22]

However, these more general discussions may actually conceal the dynamics in sectors such as metalworking, which – as noted above – encompassed an assortment of craft activities with different institutional affiliations. Building on data from the head tax records and including the whole sphere of metalworking activities in the capital 1760 and 1780, Table 2 illuminates several trends during the late-eighteenth century, not entirely consistent with ideas of decline or stagnation. The number of

Table 2. Workplaces and employees in the Stockholm metal trades, 1760 and 1780. Source: Stockholms stadsarkiv, Överståthållarämbetet för uppbördsärenden, Kamrerareexpeditionen, G1BA:21 and G1BA:24, head tax records (*mantalslängder*) Stockholm 1760 and 1780.

	Workplaces		Journeymen		Apprentices		Other empl.		Total empl.	
Parish	1760	1780	1760	1780	1760	1780	1760	1780	1760	1780
Klara/Kungsholmen	100	98	118	129	121	135	9	15	248	279
Jakob/Johannes	68	59	73	64	85	75	4	1	162	140
Ladugårdslandet	28	29	18	18	19	20	12	6	49	44
Nicolai	100	117	156	154	166	140	24	71	346	365
Maria	21	47	30	42	26	69	6	13	62	124
Katarina	39	35	40	45	34	49	3	1	77	95
Total	***356***	***385***	***435***	***452***	***451***	***488***	***58***	***107***	***944***	***1047***

Note: The numbers cover all types of metalworking businesses in the capital, regardless of institutional affiliation. Unskilled workers, like servants and day labourers, are not included (c.f. Table 1), nor are masters and manufacturers (unless they worked for someone else).

workplaces and employees increased in Nicolai parish; the former development being mainly related to a larger share of small-scale workshops, and the latter depending on the expansion of the Royal Mint, which employed some 60 workers in 1780. In Klara, the number of workplaces declined, but the number of employees increased. Metalworking weakened in the parishes of Jakob and Johannes (north of the town centre), but instead expanded in the capital's southern districts. Maria parish saw the largest increases of workplaces (particularly smaller ones) and employees. One possible explanation is that the crisis promoted a move towards small enterprises in poorer parts of the capital.[23] With that said, it must be noted that some large workplaces operated south of the town centre as well, one of which was Thomas Lewis' foundry, employing 25 workers in 1780. Based on these observations, we can conclude that there was room for the (re-)establishment of metalworking businesses in the capital, despite the crisis years of the 1760s.

Thus, while traditional scholarly perceptions of the Stockholm economy indeed have clear values, they run a risk of simplifying rather complex trends, thereby obscuring the fact that large towns had several key functions in early modern industrial development, also during crises or times of economic decline: they stimulated mobility, social encounters, and business transactions, and consequently reshaped linkages among the spheres of production, consumption, and trade. In this matter, Daniel Roche calls for an awareness of 'the capacity of the urban driving force to attract people and hasten exchanges, and so to change behaviour', while Bert De Munck and Antonella Romano stress the need to investigate 'the mechanisms with which certain types of knowledge and knowledge formation became historically entangled with the city and the urban context'.[24] In line with such remarks, there is definitely a need for in-depth qualitative perspectives on the long stagnation in Stockholm, notably so regarding craftwork and manufacturing. In this article, I emphasise three elements in particular, all of which have featured prominently in recent historical research, but largely without discussions relating to Scandinavia: (i) cross-border travelling and the transnational circulation of technical know-how and craft techniques; (ii) urban practices of mobility and training as well as of innovation, testing, and work organisation; and (iii) the various roles played by skilled craftspeople from the shop-floor world.

There is a growing ambition in European research to dig deeper into overlapping (and sometimes conflicting) 'systems of transmission of skills' as well as the multitude of 'techniques and styles' that made early modern towns into 'melting pots', using Alida Clemente's terms.[25] Workforce mobility and the dissemination of work methods are key components in this narrative. Focusing on the pre-industrial craft guilds, Stephan R. Epstein and Reinhold Reith have shown how craftsmen's movements took different shapes, depending on varying institutional environments and the degree of urbanisation in different regions.[26] Moving beyond urban guilds, thus bringing additional knowledge-making practices into the discussion, others have emphasised how state-funded artisanal mobility was linked to the development of special privilege-systems and the rise of new manufacturing industries in various parts of early modern Europe.[27] Recent studies have, moreover, accentuated how such processes incorporated an assemblage of social networks that, in turn, illuminate how the circulation and local uses of knowledge and skills depended on a variety of practices as well as 'a broad group of actors from diverse professional and geographical backgrounds'.[28]

So far, these discussions have mainly focused on densely peopled regions, such as the Low Countries and Italy, while analogous studies of Scandinavia are rarely found. In part, this uneven attention has its natural causes. The Nordic countries were sparsely populated with only a few big cities – the population in Stockholm grew from 45.000 in 1720 to 70.000 in 1770 – and with comparatively small urban sectors.[29] Nevertheless, these areas can provide important insights into the functions of town-related mobility in broader patterns of economic development, especially in terms of emphasising uncharted forms of transnational interaction or further clarifying the relations among mobile individuals, urban institutions, and cameralist administrations. For the Stockholm guilds, we know from previous investigations that craftwork practices in the capital were linked to German and Baltic towns during the entire early modern period, for example, through journeymen's tramping.[30] Other studies have highlighted how both skilled foreign experts and mobile Swedish artisans played key roles in various projects in the capital, often closely tied to the ambitions of state organs, emerging scientific institutions, or groups within the socioeconomic elite.[31] Similar observations – linking together practices of workforce mobility, processes of technology transfer, and the implementation of new *fabriques* or large-scale construction projects – have been made regarding other eighteenth-century urban settings in Scandinavia, like Copenhagen or the Suomenlinna fortress (Sveaborg) outside Helsinki.[32]

In this context, the urban metal trades are an appropriate starting point for a more comprehensive investigation of the dissemination of skills and technical know-how. Metalworking in Stockholm benefitted from well-established links to German-speaking territories and a growing interest in expanding British industries, but it also developed in tandem with the important Swedish iron production and expanding provincial manufacturing. Moreover, apart from housing a wide variety of potential consumers, including other crafts, groups of professionals, and private households, the capital also became a node for overlapping administrative and scientific ventures in the metalworking sector, as illustrated by activities that involved the powerful Bureau of Mines (*Bergskollegium*).[33] These multiple forms of interaction point to the fact that we need to add a layer of intraurban and regional transactions to the emphasis on transnational connections. Such a discussion illustrates how technical knowledge and skills with different trajectories were locally implemented, adapted, combined, and further disseminated, processes that make it possible to approach the 'circulation' of knowledge in the early modern metalworking sector in a more detailed manner.[34]

In this regard, the empirical sections of this article open the way for an in-depth understanding of important practices in the Swedish capital. They illuminate variations of training and skill formation, demonstrating how traditional forms of apprenticeship and journeymen contracts were gradually accompanied by other forms of education and employment.[35] In line with studies on eighteenth-century London and Paris, these observations can, in turn, be linked to notions of a mutable urban landscape of manufacturing, composed of a blend of large and small enterprises, but also shaped by cycles of demand, subcontracting, and collaboration that affected the sizes of workshops as well as the rhythms of workforce migration and training.[36] Relating to Maxine Berg and other British scholars, this flexibility of the urban metal trades is further proven by a continuous capacity for process and product innovation, made visible by the

countless examples of material testing, imitation, organisational rearrangement, and trials with new (and far from always successful) machines but also by a multitude of conflicts regarding privileges and production rights, all recorded in the administrative archives.[37] Taking these different elements into account, one can indeed refer to the capital as a 'bazaar-like economy', although Salvius' terms are probably even better suited: Stockholm was 'like the Heart in a Human Body', and the capital's manufacturing sector was marked by diverse and complementary forms of 'life and movement'.[38]

Linking the local dynamics of Stockholm manufacturing to wider processes of knowledge circulation, this article resists the assumption that it was the capital itself, or any specific urban group, that alone generated a capacity for creativity and change. Instead, I emphasise how knowledge 'emerged from a range of practices which are simultaneously place-bound and to be situated in broader networks [...] of people [...] or circulating objects'.[39] In doing so, however, I focus on actors who rarely feature in conventional stories about technical development in pre-industrial Sweden, particularly artisans. There are good reasons to do so also when taking a wider research context into account. While workforce mobility in the European metal trades has been the subject of several studies, there is a lot more to be said about the roles of 'mobile artisans' – in this article, mainly those who were active in the state-supported manufacturing trades – in processes of innovation, technology transfer, and the codification of techniques.[40] One can refer to many of the actors studied here in terms of intermediaries or go-betweens, clarifying how they used 'embodied competences' to articulate 'relationships between disparate worlds or cultures'.[41] Still, as noted by David Garrioch, such markers must not be used at the expense of also highlighting the simultaneous and multifaceted roles of artisans as 'active participants in the creation of skills and knowledge'.[42]

Consistent with views put forward by scholars such as Liliane Hilaire-Pérez and Paola Bertucci, I show how metalworking practitioners in the Swedish capital made key contributions to the circulation of knowledge and skills; they tested new procedures, collaborated in technical matters, and shared their know-how with colleagues, while also promoting the utility of materials, devices, and work methods to local authorities and state bodies. Thus, they played essential parts in transnational processes of codification, often in fairly open ways and not seldom in close interaction with civil servants, merchants, and scholarly experts.[43] These notions of a multifaceted artisanal 'milieu technique' are important when seeking to throw new light on urban knowledge-making, nuancing the ideas of de-industrialisation and stagnation in early modern Stockholm.[44] By showing how this context of manufacturing was shaped by 'bottom-up circulations, and their entanglements or disjunctures with those planned and created by the economic and political elites', this approach moreover answers the call to reintroduce craft skills and artisanal work in discussions of technical and economic improvements, administrative alterations, and epistemological developments during the first industrial revolution.[45] Accordingly, in order to get to grips with the crisis of the 1760s and its effects on the Stockholm metal trades, we must take our departure not in the crisis itself, but rather in a solid understanding of the urban context, the knowledge-making practices, and the working people that the crisis affected in various ways.

Mobile artisans and mid-century expansion

The manufacturing-oriented agenda from the 1730s brought about a growing interest in the spread of useful know-how, with state efforts to recruit skilled labour and technical expertise, fund study tours, and reimburse the introduction of new work methods. Studies dealing with the undertakings of civil servants involved in the metal trades have shown that such projects stretched far beyond the borders of the Swedish realm, including an assortment of mines, manufacturing communities, and towns, as well as an intricate mix of 'espionage' and 'intelligent travel', using Bertucci's terms.[46] To these observations must be added the fact that the cross-border circulation of knowledge involved various other transnational groups – including merchants, scholars, diplomats, and foreign manufacturers – who offered services regarding transport, financing, and the mediation of contacts.[47] Artisans' roles in these processes have been highlighted to a much lesser extent. As shown here, European journeys provided opportunities for skilled craftsmen (of both Swedish and foreign origin) to shape the conditions for knowledge circulation, often through diverse contacts with patrons, local intermediaries, and state bodies. Tracing the itineraries of metalworking artisans moreover enables a focus on associated practices in the Stockholm trades, highlighting both the 'politicization of technology' and the making of 'open technique', relating to Hilaire-Pérez's investigations.[48] Put differently, it uncovers the ways in which cross-border travelling was entangled, on the one hand, with administrative intervention (through funding, privileges, and supervision), and, on the other, with varying forms of skill transfer, collaboration, innovation, and – not to forget – conflict.

In line with the protectionist policies, state-supported mobility often targeted trades considered as particularly useful for the Swedish economy. The making of instruments and fine tools came to enjoy a special care from the 1750s, as a response to demand from learned professions, precision crafts, and the military. When instrument maker Jacob Christopher Ketscher travelled in France and England, visiting Parisian workshops as well as British '*fabrique*-towns', his endeavours were monitored by *Kommerskollegium* and the War Office (*Krigskollegium*) through correspondence with on-site agents such as London envoy Arnold Wynantz and merchants Abraham Spalding and Gustavus Brander.[49] Similar tours were made by makers of optical and mathematical instruments, likewise depending on patriotic networks that extended from state bureaus and the Royal Swedish Academy of Sciences (*Kungliga Vetenskapsakademien*) to trade offices, consulates, and academies in other European cities.[50] Such efforts could bring about enhanced possibilities for observation and training. When returning to Stockholm, in 1756, Ketscher had practised the making of surgical tools, scissors, and razors, but also gained knowledge about steel processing and polishing. State supervisor Schröder assisted in equipping a two-floor workshop in the town centre, and the journeyman was soon also further supported by a special 'establishment premium' from the state funds.[51]

On many occasions, however, the influx of know-how was contingent on inadvertent contacts made on the move. Travelling abroad on his trade in the late 1730s, Matthias Holmerus abandoned a career as a jeweller after being urged by the prospective Royal printer, Peter Momma, to instead make inquiries – in Paris, London, and likely also in Dutch cities – into the 'more useful and less cultivated' production of metal types. Back in Stockholm, and in cooperation with Momma, he

set up a brand-new type foundry and managed to get hold of a twelve-year exclusive privilege to develop his craft, having outwitted a potential competitor.[52] Other cases further clarify how mobile craftsmen were incentivised to travel home, so that valuable skills could be spread in the domestic trades. Foundry worker Anders Dahlström was hired in Paris, in 1746 or 1747, by touring architect Carl Hårleman, who was in search of artisans for the completion of the Stockholm castle. Dahlström had by then spent three years training in the French capital, mainly with one 'skilled master'. After the return to Sweden, he was further educated by French workers, later recalling that he had paid two of his foreign colleagues 600 copperdaler for them to share their skills in brass work and gilding. Eventually, he received a similar reward as Ketscher to create a workplace of his own in the capital.[53]

Ketscher, Holmerus, and Dahlström thus relied on subsidies, special privileges, and other forms of administrative support to set up their urban enterprises, forms of assistance that also set them apart from guild craftsmen, or (in Holmerus' case) other aspiring manufacturers.[54] It must, however, be noted that these mobile artisans came to operate their workshops on a small scale, within the frame of a traditional – albeit expansive – craft economy. Holmerus' type foundry, in Klara parish on Norrmalm, employed a handful of workers in the 1740s. Parting with Momma, to instead focus on making types for the Bank of Sweden (*Riksens Ständers Bank*), Holmerus kept a small *fabrique* in the same neighbourhood during the following decade.[55] Dahlström, in turn, erected his foundry workshops in the quarter Uppvaktaren, also in Klara parish. Like Holmerus, he depended on a small group of employees, working with a few apprentices and journeymen, among whom the tasks of casting, filing, and chasing were divided.[56]

Foreign journeys were, however, not only a way for craftsmen to get their hands on privileges, loans, and rewards. They could also bring about possibilities to perform pivotal intermediary undertakings, or actively shape the conditions for skill transfer, future labour relations, and the potentials for implementing new work methods. Some travellers, like watchmaker Johan Lindquist, not only procured useful know-how abroad but they likewise succeeded in luring colleagues to follow them to Stockholm. Lindquist trained for some years during the 1750s with the famous Julien le Roy in Paris, acquiring his employer's 'invented method of making repeater pocketwatches'. Still, of equal importance to Lindquist's prospective business was that he recruited five journeymen during his stay in the French capital. In 1754, he was paid 600 *livres* by *Manufakturkontoret*, intended to cover all expenses for him and his new workers on their journey to Sweden.[57] One of the hired men, Swiss Augustin Bourdillon, later stressed that he had been 'persuaded' to travel north because he 'thoroughly knew the art of [making] repeater watches'. This esteemed know-how in the mechanical arts played a decisive role when Bourdillon eventually set up a Stockholm-business of his own.[58] As we will see, such watchmaking enterprises are also clear examples of businesses deviating from the predominant pattern of small-scale manufacturing.

Similar roles can be ascribed to manufacturers in Stockholm, using their wide networks to bring about staff recruitments and skill transfers. One example is the introduction of the French *argent haché*-technique during the 1750s. As one of the pioneers, Royal chaser Simon Pantaleon – himself recruited to Sweden in 1745 with his father, but then further educated in Paris – relied on hiring one 'argenteur' from his native country. His competitor, merchant Jean Crés, made sure to employ no less than 'five special workers'

for the making of silvered brassware. Among them was French master 'S:r la Cour', one of the artisans who had previously been paid to instruct Anders Dahlström.[59] Comparable arrangements took shape in other crafts. Likewise using his French connections, tinplate maker Nicolas Joseph Isbeque hired two Parisian journeymen in 1755 for the making of *quincaillerie* and 'Chinese-style leadwork'.[60]

Artisanal immigration was, however, never a straightforward business. Many Swedish craftsmen working abroad did not see a move back home as an adequate alternative, despite promised state support. Visiting Dresden in the early 1750s, during his foreign study tour, Schröder learned that skilled Swedes working in the Saxon capital were not motivated to 'bring the art back to their fatherland', as current working conditions offered better 'communications' to urban markets and numerous courts. Moreover, the emigrant craftsmen felt 'less restricted' by regulations and guild statutes, compared to the situation in Swedish towns.[61] Among the manufacturers who did return to Stockholm from other European cities at mid-century, quite a few became involved in disputes with other – often guild-affiliated – craftsmen about rights of production. When former goldsmith journeyman Hartwig Losch came back to the capital in 1755, having worked several years in both London and Paris, *Kommerskollegium* saw no problems in granting him permission to set up a workshop for the production of goods made from copper-based alloys such as Pinchbeck and Tombak. He was, however, strictly forbidden to interfere with the exclusive rights of the goldsmiths' guild by making pieces of gold and silver.[62] Needless to say, there was a significant potential for cheating and conflict regarding such officially stipulated boundaries.

The situation was often no easier for foreign craftsmen, some of whom shared the fate of Mr. Rochebot, a French steel polisher arriving from Berlin in 1754 with good references. Schröder tried in vain to get the Frenchman a job with one of the capital's sword makers or gunsmiths, and later stated that it was likewise pointless to let him work at any water-powered metal works, as Rochebot only knew 'manual polishing'.[63] The hiring of workmen from the outside could also result in conflicts in need of administrative intervention. Soon after starting up his business, manufacturer Isbeque put himself in debt and had to cancel the salaries to his contracted French journeymen, Jean Raymond Laurier and Gideon Lavangie. The latter asserted their right to part with the employer and demanded compensation for non-paid wages as well as promised money for the journey back to France, a dispute ending at the *Hallrätt* with a positive ruling for the two workers. In order to prevent them from leaving the realm, Schröder arranged with new jobs at the large Forsmark iron works in Uppland.[64]

Making the move to Forsmark, Laurier and Lavangie became part of an intricate national pattern of workforce migration, in which the capital formed a hub for the circulation of technical know-how. To nuance the, to some extent correct, assumption that early modern towns relied on a continuous inflow of skilled individuals, it is thus important to relate foreign journeys and transnational transfers to regional and local forms of mobility.[65] In the case of the Stockholm metal trades, it is evident that the urban context attracted artisans – masters as well as apprentices – from the Swedish provinces. At the same time, the capital served as a vital training ground, providing other towns and rural manufactories with accomplished workers of various professions. In many cases, such movements were linked both to the businesses of European travellers and to state ambitions to support the copying of foreign-made metal items.[66]

The relocation of provincial makers to the metropolis was often preceded by collective investigations and displays of skills, activities that not only provided a basis for the implementation of utility-oriented policies, but which were also key to the codification and wider circulation of practical know-how.[67] However, the underlying strategies of the artisans involved took different forms. Watchmaker Petter Ernst likely intended to obtain a generous 'invention premium', rather than to leave his home in Växjö (Småland), when exhibiting his chronometers for King Adolph Frederick in 1752. Once the 'machines' had been 'scrutinised' by *Vetenskapsakademien*, however, voices were raised that it was better if Ernst elaborated on his technical projects in Stockholm. Academy member Jacob Faggot vouched for Ernst's 'genius and skill', stressing that he possessed an 'inventive gift' comparable to 'the best *Mechanicus*'. *Manufakturkontoret* agreed, and it was decided to offer Ernst several rewards, on condition that he took on the management of a late colleague's *fabrique* in the capital. The following year, he settled in his new hometown to start production.[68]

Intersecting elements of artisanal enterprise and authority-based handpicking are also evident in other cases related to the entries of provincial metalworkers. Like Ernst, smith Olof Beckelin, from Lidköping in Västergötland, was recognised as contributing to the spread of useful know-how, after being given a chance, in 1758, to present samples of watch springs, chains, and fine wire to Schröder. The latter promptly organised tests with Beckelin's produce, involving potential users such as Johan Lindquist and Petter Ernst. The watchmakers were content, later asserting that Beckelin, 'by his own endeavour and invention', had advanced the making of watch- and clock details to the same level as abroad.[69] The newcomer was awarded a privilege to set up a Stockholm workshop, and soon expanded his business. He made springs and chains to be fitted into the mechanical curiosities of men like Ernst (see Figure 1), and thus became a key player in the subcontracting network that supplied the capital's watch- and clock-makers, while also experimenting with the making of previously imported steel tools.[70]

The creation of workshops in the urban metal trades was likewise linked to the education of a growing number of trainees, many of whom came from provincial communities, and several manufacturers were offered 'training premiums', intended to cover the instructional costs.[71] Returning from their foreign tours, Jacob C. Ketscher and Anders Dahlström both took on the subsidised training of apprentices. Still, they were also involved in other types of skill transfer. In 1758, Ketscher was paid to share his know-how in sharpening and polishing for six months with a cutler from a large manufactory, and the following year, on Schröder's request, he hired a journeyman from the Duvnäs *fabrique*, south of Stockholm, to teach him the making of scissors.[72] Dahlström, in turn, stressed similar achievements, mentioning several journeymen who had 'perfected themselves in the art' in his workshops. The two-year education of 'student' Eric Wellenius, in 'everything that belongs to the casting of brass and metals', was fully paid for by merchant Jean Lefebure. Leaving Dahlström, in 1758, Wellenius was soon hired to supervise the casting at Lefebure's brass works in Norrköping.[73]

Such overlapping cycles of training were a key component of the positive momenta that characterised some trades during the 1750s and early 1760s. Toolmaker Anders



Figure 1. Carriage clock, made by Petter Ernst, second half of the eighteenth century. Source: Nordiska museet, NM.0077307A – C. Photo: Ridde Johansson, Nordiska museet, Stockholm (the image has been slightly cropped). The subcontracting network that supplied Stockholm's watch- and clockmakers expanded during the mid-1700s, and manufacturers such as Olof Beckelin specialised in making parts, fine tools, or equipment to be used in the watchmaking workshops. Reproduced with permission from Nordiska museet.

Wahlbom, himself educated both at a rural manufactory and in a prominent urban instrument-making workshop, was one of the artisans who expanded his business during this period. Credited for offering perfect substitutes to imported sets of fine tools, he also taught a new generation of steel-ware producers, as paid for by state grants. In 1757, when Schröder visited the workshop on Norrmalm, Wahlbom had recently hired five trainees, a number that more than tripled over the following decade. In 1766, he engaged 17 workers at his new *fabrique*.[74] At that time, the only manufacturer in the capital's metal trades with a larger workforce was Johan Lindquist's former journeyman Augustin Bourdillon, whose watchmaking workshops employed close to 30 workers. Not favoured by any generous rewards, Bourdillon nevertheless later stressed that the teaching of 'Swedish children' had been supplemented by the contracting of skilled foreigners, for tasks such as enamelling and gilding, colleagues whom he had also paid rather large sums for them to share some of their expert know-how.[75]

There were thus a variety of training solutions outside the guilds, even though skill transfer in the manufacturing trades in most cases still relied on traditional apprenticeships. In line with observations from other early modern European cities, this diversifying landscape of knowledge formation also included additional institutions.[76] In 1762, after completing a state-funded training with French sculptor Jacques Adrien Masreliez, the draftsman in *Vetenskapsakademien*, Olof Årre, began teaching metal-working youngsters at a new drawing and modelling 'school'. Schröder, who of course had a hand in the project, noted that Årre instructed 16 apprentices and journeymen 'from various professions' during the following two years. Between 1765 and 1769, he had ten boys in

training, including one of Bourdillon's enamelling apprentices.[77] This example points to the growing interest from both political organs and scientific institutions in the dissemination and codification of craft methods. As we will see, administrative experts indeed came to take on more active roles in such processes during the following decades, although they continued to depend on a combination of labour- and education-related efforts made by skilled artisans on the shop floor.

The urban assemblage of mobile people and training practices played a vital part in nurturing imitation, experimentation, and new forms of collaboration, processes that both reflected cameralist regulations promoting import substitution and, at the same time, provided a basis for innovative achievements and commercialisation. In this way, the capital's metal trades exhibit parallels with developments in other European – notably so British – cities.[78] The making of instruments and tools, to mention one good example, often relied on joint efforts related to the circulation and innovative copying of foreign artefacts. Manufacturers Ketscher and Wahlbom were both selected, by Schröder, to take part in the attempts to bring about a stable production of surgical equipment. One of the intended users, surgeon Olof af Acrel at the Stockholm Seraphim hospital (*est.* 1752), provided the artisans with French-made 'models' from his private collection. The project made a clear impression on Ketscher's production, and in 1758, he provided *Krigskollegium* with eight field kits – the first ones made in Sweden, according to Schröder. Wahlbom was hired for yet other types of imitation, making copies of English precision tools that were delivered to the capital's watchmakers or put up for sale in a shop at the busy street Storkyrkobrinken.[79] The shopkeeper, merchant Joseph Schürer, also sold items made by Anders Schagerborg, one of several trinket makers who profited from the state-aided copying of tools and cutlery. Equipping knives and instruments with stylish handles and casings, he created products that were 'if not on par with, then at least fairly similar to the French', in Schröder's opinion.[80]

The imitation of foreign goods was combined with the trialling of new metal assortments, often sent from provincial iron works and manufactories to be evaluated in the capital. Schröder consulted skilled men like Ketscher and Wahlbom when it came to the prioritised refinement of steel, and, at times, he made sure to be present himself when the qualities of different steel sorts were exposed during workshop-based tests.[81] Other artisans instead actively participated in the creation of metals and mixtures, illustrating the links between the making of 'tangible things' and the creation of 'generalized, replicable, and transmissible knowledge about nature', in line with Pamela Smith's discussion.[82] Not officially stated as an originator – which would have given him an opportunity to receive an 'invention premium' – foundry worker Dahlström was nevertheless deeply involved in composing one of several new copper-based alloys that attracted the attention of civil servants and learned men. In September 1757, Schröder even made his way to Dahlström's foundry to observe the melting process firsthand, only to conclude that the 'red metal compound' hardly was any 'new invention'.[83] Olof Beckelin succeeded better with his ambitious project to make blister (or cementation) steel within the frame of his urban business. The erection of a small furnace on Södermalm – in a 'new, but mostly English, manner' – was paid for by merchants Claes and Johan Abraham Grill, who also had let Beckelin prepare pieces of bar iron at their iron works in Österby (Uppland). In return, Beckelin promised to share his vast know-how in iron- and steelmaking with the smiths at Österby. In July 1760, he

presented samples of a new blister steel, made from Österby iron, to Schröder, who arranged a test procedure. Artisans in the capital and the smiths at one provincial *fabrique* made cutting tools from the steel, and the reviews were highly positive as the material was 'compact' in processing and of a 'more refined' quality than other Swedish brands.[84]

Yet others, like Petter Ernst, even managed to promote innovative accomplishments outside their main branch of business, to the recognition of both scientific authorities and utility-oriented state bodies. After 'over nine years of work' and 'countless' experiments, Ernst was able to present his own method of making Verdigris – green pigment prepared by treating copper with acetic acid – entirely from 'Swedish substances and products'. Receiving a ten-year exclusive privilege for production in 1760, he described the entire process to *Manufakturkontoret*, and stressed the 'meticulousness and deeper knowledge' as well as the sound practical experience required to achieve 'the detailed principles of chemistry and nature'.[85]

As indicated above, there was indeed room for conflict in this context of copying and material innovation, not least when the work of outsiders infringed on guild privileges. These 'horizontal' disputes highlight the intricate boundaries between 'improvement, imitation, and theft' in the urban trades.[86] They were not restricted to the making of luxury goods, but also arose regarding mundane consumer items. Copper manufacturer Petter Winberg, yet another London-returnee, faced a resolute opposition from both guildsmen and local authorities when promoting his business in 'enamelled or brown copper'. In 1764, he even saw his marketed products being seized by the town's public prosecutor (*stadsfiskal*), as some of the pots, pans, and kettles were found to be 'ordinary and polished' wares, the making of which the capital's coppersmiths claimed exclusive rights. The conflict, involving several authorities, lasted several years before a solution was finally reached through *Kommerskollegium*'s intervention.[87]

Disagreement was an inevitable element in a privilege-based urban economy where work methods and novel forms of technical know-how were continuously introduced and adapted. Still, by promoting various forms of testing, imitation, collaboration, and product innovation, these processes also gave rise to intensified interaction among makers, traders, learned men, and officials, links that, in turn, were essential in favouring commercial growth as well as the circulation and codification of useful techniques. In some cases, and similar to the remarks made by Ken Alder, cross-border knowledge circulation and imitative ambitions also created a basis for process innovations in the form of technical and organisational alterations.[88] This is clearly shown by the growing number of workplaces that relied on a piecework-model according to 'the English way'.[89] In the watchmaking trade, subcontracting was accompanied by an elaborate internal division of labour at a handful of *fabriques*, where each of the workers practised 'fairly uniform skills', as put by Schröder in 1760. Johan Lindquist employed nine journeymen and four apprentices, among whom work was divided so that some only made unique parts and others worked with finishing. Besides promoting 'the same quality and price' that were achieved at foreign *fabrique*-establishments, such a method brought about 'a greater dexterity', according to the state supervisor, and it was enhanced by the installation of technical gear and efforts to improve specific manual operations.[90] Analogous plans were launched in other trades. In 1766, when granting Anders Wahlbom a large loan, one of the Diet's

delegacies noted that the toolmaker, notwithstanding the quality goods he already made, would be helped by equipment for the 'improvement' and 'shortening' of the work process. Three years later, Schröder noted that Wahlbom had initiated such upgrades in his workshops, installing equipment such as stamps and dies as well as arranging for a new sharpening- and polishing mill.[91]

Adding these observations to the other elements highlighted in this section, it can well be said that Wahlbom's *fabrique* was a nexus for activities that characterised the mid-eighteenth-century expansion for metal manufacturing in Stockholm. It was shaped by imitative ventures, collective testing, and new forms of training and codification, as well as by organisational and technical alterations, all made within a framework of small-scale craft production. This development incorporated numerous overlapping movements, ranging from European travels and cross-border transfers of useful know-how to regional and local workforce mobility. Administrative regulation and elite plans indeed played important parts in these processes of circulation, but I have also emphasised the 'bottom-up' strategies involved in the making of 'open technique', especially by focusing on mobile artisans' endeavours to bring advances about in their respective businesses. Moving between individual workshops, the local dynamics of a privilege-based economy, and the wider trajectories of people, methods, and skills, the discussion above has thus aimed to highlight different, but often intricately entangled, practices of work and knowledge-making, and, in so doing, to create a more 'spatially aware approach' towards the everyday creativity and mobility of the urban metal trades.[92] Such a perspective constitutes a good basis for elaborating a better understanding of the development during and after the turbulent 1760s, and, on a more general level, for bringing nuance to the idea of a long stagnation.

Runaways and the rise of new metalworking enterprises

Whilst Wahlbom improved on the work process at his *fabrique*, the positive trend had turned into an ever more uncertain situation due to the tangible economic and political disorder. Many of those not benefitted by the same generosity as the toolmaker were forced to seek out better conditions elsewhere; some left for the provinces, others looked farther afield. Still, in the wake of the difficult years, there are also examples of new innovative projects, although often presented – if at all – as rare exceptions that confirm a general trend towards stagnation in the capital. In contrast, I argue that these activities put focus on the continuously dynamic function of urban manufacturing in the gradual move towards an industrial society. This development included changing patterns of transnational knowledge-gathering and increased focus on the technical aspects of production as well as plans for new educational institutions and early measures in support of liberalisation. In all these respects, imperative roles were played by a group of entrepreneurial administrators, who used their 'manufacturing expertise' to mediate between political arenas, scientific settings, and metalworking practices.[93] Still – and considering that a large part of the training, technology transfer, and codification of techniques was persistently tied to a mutable context of small-scale craft production – we must also recognise the key positions held by mobile artisans. They not only moved between Stockholm and other communities but also creatively integrated material transformations, knowledge circulation, and the exploration of new markets, thus actively generating modifications in the metal trades.

In late 1764, Schröder drew attention to problems faced by the capital's manufacturers, a number of whom had closed their workshops during the preceding years. Among them were the abovementioned Nicolas J. Isbeque, Simon Pantaleon, and Jean Crés. The negative trend spread during the mid-1760s, with several artisans leaving for rival nations. French *argent haché*-manufacturer Fredrich T. Lemair and maker of mathematical instruments Johan Ahl moved to Denmark, while watchmakers Jean Fazie and Petter Norsten, metalworkers Eric Norgren and Johan Falk Ericsson, and file maker Johan Sundsten all made their way to Russia.[94] The exodus also included former state protégés. Likewise attracted by the industrial projects launched by Empress Catherine II, Jacob C. Ketscher left Stockholm already in 1763, having failed to continue his increasingly indebted business. The instrument maker travelled via Berlin to Saint Petersburg, where he came to produce surgical tools for the Russian army.[95]

These movements are clear signs of the combined process of economic decline and political reorientation that characterised the manufacturing sector during the 1760s. As Schröder later would emphasise, it was certainly bad that workers had been 'scattered all over the Provinces and undertaken other livelihoods', but still worse that a substantial number of people had 'left the Kingdom', to practise their crafts 'with better success and profit' abroad. The negative impact was twofold, according to the state supervisor; skilled artisans, whose training had been publicly sponsored, were 'missing from Society at the moment when the greatest benefit could be expected from them', while neighbouring kingdoms could instead 'improve at our expense'.[96] However, it must be noted that artisanal mobility during the crisis was not limited to one-way transfers. Illuminating some examples in-depth will add clarity into the intricacies of failing projects as well as the complex geographies of work and training.

While the funds for artisanal study tours became more limited, there are indeed evidences of new state-supported voyages, although not all of them ended up benefitting the Swedish metal trades. In 1765, journeyman Georg Niclas Carsberg was persuaded, by Schröder, to make investigations into the production of cutting steel items in England. Suitably enough, he had already acquired skills in the said trade, training with Jacob C. Ketscher and then working for, among others, Anders Wahlbom. The same year, Carsberg left for London, where he came to practise the making of razors and scissors. When returning one year later he had trained with 'skilled manufacturers', to the authorities' contentment, and funding was raised for the journeyman to set up a workshop in Stockholm. Due to the harsh economic conditions, however, it was soon decided that Carsberg should travel back to England to gain more experience. He departed again in 1767, and this time he remained overseas. Some ten years later, the discussion about supporting him to move home gained a new momentum, related to ambitions to reinforce the production of surgical gear. *Kommerskollegium* and *Krigskollegium* were not too optimistic, noting that Carsberg had a 'more advantageous establishment in England'. Besides, there were other manufacturers in Stockholm who met the requirements.[97] The former state-funded traveller instead settled for good in London, managing an instrument-making business on Great Windmill Street.[98]

Others only left Stockholm temporarily to escape acute problems. The increasing number of bankruptcies and financially conditioned getaways was one key motive behind the revised bankruptcy laws of 1766 and 1773, which offered legal rights to the growing – and largely unprotected – group of debtors.[99] These changes came

too late for Olof Beckelin who, in 1765, had to evade his creditors' harsh demands by fleeing to Norway. Still, he took the opportunity to return later the same year, when offered a possibility to settle his debts and resume his business. Good references from the capital's watchmakers promoted the re-entrance, but a more imperative reason was probably the authorities' fear of losing Beckelin's 'rare genius and skills' to any competing nation, as had happened in the case of Jacob C. Ketscher.[100]

Movements forth and back between Stockholm workshops, provincial manufactories, and work practices in other European towns during the economic downturn could in fact also be one way for artisans to improve on their skills and shape their careers through diverse work experiences and the active use of contact networks. When tinplate worker Jean R. Laurier, one of Nicolas J. Isbeque's old employees, re-appeared in the capital in 1768, he could look back on five years as a traveller. In 1763, he had left Forsmark iron works for Paris, only to return some two years later, having informed Schröder about his desire to escort a compatriot – the inventor of an 'approved art of putting lacquer or amalgam on copper, brass, and iron wares' – to Sweden. It is unclear if this colleague joined the journey northwards, but Laurier showed up together with his family. In April 1765, they made a short stop in Stockholm on the way to Skinnskatteberg in Västmanland, where Laurier had been promised a supervisory position at a tinplate works. When production ceased three years later, amid the crisis, he once more found himself on the move, making his way to Saint Petersburg to 'try his luck' together with one of his former employees. Again, it did not take long before he returned. In October 1768 – twelve years after being hired by Isbeque – he finally settled with his family in Stockholm. A later advertisement stated that 'all sorts of tinned Iron wares' could be bought in Laurier's *fabrique* at the street Bryggargatan, a business continued by Widow Sara Laurier after her husband's death in 1776 or 1777.[101]

Despite the harsh times, Laurier was not alone in starting up a metalworking business in the capital during the latter half of the 1760s. The removal of obstructive regulations – notably so related to the 'freedom' of young craftsmen to practise their trade independently – paved the way for new enterprises, and in 1769, Schröder counted some 20 workshops founded during the preceding five years, covering various crafts.[102] At the same time, a tightening of state funding and the plans for Eskilstuna *fristad* show that the increased liberty did not extend to all metalworking practitioners, nor was it equivalent to reduced state interventionism.[103] Steel-ware makers Göran Tawonius and Petter Åberg, two of Anders Wahlbom's former trainees, were both denied support during the Diet of 1771–72. Tawonius tried to justify his project by referring to work experiences 'both in this country and abroad', but Schröder noted that the applicant's poverty would hamper the business; it was better if men like Tawonius were hired by others. Åberg received a similar reply, when applying for a reward for two invented machines for the 'improvement of the work method'. In this case, *Kommerskollegium* even noted that expenses on small workplaces in the capital could 'impede the intended purpose' with the *fristad*.[104] The latter establishment, endowed with ample waterpower and large mechanical facilities, expanded during the 1770s. In contrast to Stockholm, moreover, it was not marked by 'the discord and jealousy' of an old privilege order, as put by Schröder, in which guildsmen and manufacturers 'time after time' had been granted 'different privileges, benefits, and freedoms [...] in the same trades'.[105]

Notwithstanding these disadvantages, new investments materialised in the capital, as proven by examples from both heavier metalwork and detail manufacturing, and to an increasing extent they were inspired by the expanding British metal trades. With a proven capability from Carron iron works in Stirlingshire as well as from Johan Cahman's foundry in Gothenburg, Thomas Lewis had no problems in raising advocates for his project at Bergsund. As noted by Schröder and his colleague Samuel Sandels, in 1769, the casting technique, by which pig iron was melted in reverberatory furnaces, was highly promising as 'discarded canons' and 'broken castings' could be re-used for new products. Thanks to his connection with the wealthy merchant Robert Finlay, Lewis could secure loans from both *Kommerskollegium* and the Iron Masters' Association (*Jernkontoret*) and expand his business.[106] In 1771, potential customers were informed about pots, kettles, and newly 'invented' stove dampers from 'Tom's foundry' that could be purchased in a shop at the street Lilla Nygatan or by ordering.[107]

The same year, everything seemed to be lost, as Finlay went bankrupt, thus also putting the foundry in danger. In October 1772, also Lewis had to apply for bankruptcy, but again succeeded to gain the authorities' support. *Kommerskollegium* and *Jernkontoret* enabled the continuation of production, and when a new merchant firm was involved three years later, the business was definitely saved.[108] Lewis could now devote more attention to technical elements, and, in so doing, he looked towards both an international context of metalworking and the prospects of the Swedish market. He hired skilled experts from his native country to handle supervisory duties at Bergsund and enlarged the foundry with new facilities, all built according to 'the English way'.[109] Through his British contacts, Lewis also gained access to an arena for exchanging technical know-how. He followed the introduction of 'Mss:rs Boulton and Watt's new invented Steam Engines' at the Carron works, but also informed his old employers about Swedish techniques such as the 'properest method' to make cannons 'cast solid and afterwards bored up', as put by Carron man Charles Cascoigne in 1775. Moreover, in order to provide the foundry with suitable pig iron, Lewis bought a blast furnace of his own in Nyhyttan (Västmanland), a facility that became the site for additional technical experiments.[110] The successor at Bergsund, merchant Daniel Asplund, built on these achievements when further expanding the company after Lewis' death in 1783 (see Figure 2).[111]

The Bergsund foundry is just one in a series of, more or less successful, attempts with new production methods in the Stockholm metal trades during the late 1700s and the early nineteenth century, inspired by British models.[112] A common theme is the growing focus on the technical elements of production – with improved furnaces and novel machinery – together with an emphasis on a more elaborate division of labour. Such initiatives called for inclusive and process-oriented competences, largely to be found within a group of mobile civil servants who possessed a combination of academic education, administrative training, and worksite experience. Recognising the links to practices of codification, intelligent travel, and supervision that had engaged officials like Schröder, and others before him, one could nevertheless speak of these projects in terms of an accelerating formation of 'manufacturing expertise' that made technology into a matter of systematic administrative knowledge production, while at the same time merging cameralist regulation with a private sphere of commercial interests.[113]



Figure 2. *Utsigt af Bergsund 1796*, J.G. Meijer, 1796. Source: Stadsmuseet i Stockholm, SSM 11751:0 (the image has been slightly cropped). Located on the southern outskirts of Stockholm, as clearly shown on Meijer's drawing, the Bergsund foundry grew into one of the most important metalworking enterprises in the capital during the late-1700s. Reproduced with permission from Stadsmuseet i Stockholm.

One of those who showed an interest in Lewis' innovative business was Bengt Qvist Andersson. Succeeding Schröder as supervisor for fine metalworking, in 1771, he also developed a personal commitment to the application of British technology in the urban metal trades. Having acquired useful know-how about steelmaking methods during a tour in England 1766–67, Qvist founded a steel works in Ersta on Södermalm, for the 'remelting of ordinary blister steel' or, in other words, the making of crucible steel. In 1769, he gave a detailed outline of the production, stressing that it relied on the construction of proper coal-fuelled air furnaces and the use of crucibles made from 'the best Sturbridge clay'.[114] Supported by funds from *Jernkontoret*, it nevertheless took him a decade of modifications before all technical components interacted as desired. During the 1780s, batches of steel were sent to both Swedish and foreign manufacturers to be tested, and the reactions were positive: Stockholm watchmaker Eric Lindgren reported that the material 'in all respects corresponds to the best hitherto known English steel sorts', while the renowned French cutler Jean-Jacques Perret stressed that it was 'of a good grain, clean, fine, [and] homogenous' when worked upon.[115]

Although Qvist's crucible steel was used for the making of both tools and fine cutlery, the steelmaking enterprise never took off commercially. As noted by the official himself in 1787, the Swedish market had 'too few hands' (meaning: a limited number of potential users) to balance the production costs. The Ersta steel works languished and finally closed in 1796.[116] Still, the innovative accomplishments seem to have inspired others with ambitions to establish in Stockholm. Sharing Qvist's experience of English

metalworking, more specifically from studying the making of mathematical and optical instruments in London, former navy engineer Charles Apelquist set up a *fabrique* with 'mechanical facilities' on Kungsholmen in 1785, a business that has been referred to as the first mechanical workshop enterprise (or *mekanisk verkstad*) in Sweden.[117]

Apelquist's business was rooted in a traditional privilege-based economy, benefitting from exclusive rights and state funding, but, just as in the cases of Lewis and Qvist, he indeed set the stage for something different. In the late 1780s, his *fabrique* or 'Factory' (*Factorie*) employed close to 40 workers who made 'iron- and steel tools in the English way' as well as 'mechanical machines'. Together with his colleague, he also incorporated previously imported goods, like screw jacks and wood screws, into the product assortment, made by the combined use of manual labour and technical gear, and targeting novel segments of the Swedish market.[118] The enterprise was successively expanded to include yet other branches of production. In 1794, Apelquist bought an old gun foundry on Norrmalm, only to move the production to new facilities in Marieberg, where both cannons and textile machines came to be produced. In 1809, he relocated all his other workshops to Marieberg, and the 'Factory' on Kungsholmen was instead taken over by British manufacturer Samuel Owen, another technical pioneer and founder of a large metal works for the construction of machines and casted items.[119]

Among the entrepreneurs at the turn of the century was also Gustaf Broling, an employee in *Bergskollegium* who, just like Apelquist and Qvist, had acquired valuable experiences during a study tour in Britain 1797–99. Combining 'a rare perseverance and diligence with beautiful knowledges in chemistry and mechanics as well as other useful sciences and arts', as noted by *Kommerskollegium*, he set up a metalworking business in Stockholm after his return; a workshop for surgical gear was followed by one – more successful in terms of 'seeking the consumer' – for 'Jappaned iron ware' (or lacquerware). In 1801, these worksites together employed some 20 'in-house' workers. Broling continued his project by the erection of a crucible steel works, housing 'three smelting furnaces and one forging device, powered by a small steam engine' (see Figure 3). In this case, he was even credited with having scrutinised William Huntsman's famous workshops in Attercliffe 'with an investigator's eyes', a visit that certainly served as an important example for the steelmaking enterprise in Stockholm, although a localised practical know-how had also taken shape through Qvist's earlier business venture.[120]

Like some of his predecessors, Broling pointed to challenges for further expansion, stressing both the small national market and a lack of trained workers. The 'maintenance and improvement of useful crafts' still depended on benevolent encouragements from the Swedish state.[121] Nevertheless, together with men like Qvist and Apelquist, he also pushed on in novel directions. As noted above, attempts to improve work methods with references to 'the English way' were made in the capital already during the mid-1700s, but experts at the turn of the century made more comprehensive efforts in this area, shaping their enterprises through a combination of organisational arrangements, installations of machines, and creative adaptations to changing demands. As I have shown elsewhere, this development also incorporated intensified links among product innovation, consumer preferences, and commercial practices in the capital. At the same time, increasing international competition prompted investigations into the combined commercial and technological challenges of Swedish metal production.[122] In the latter area, one solution was thought to reside in

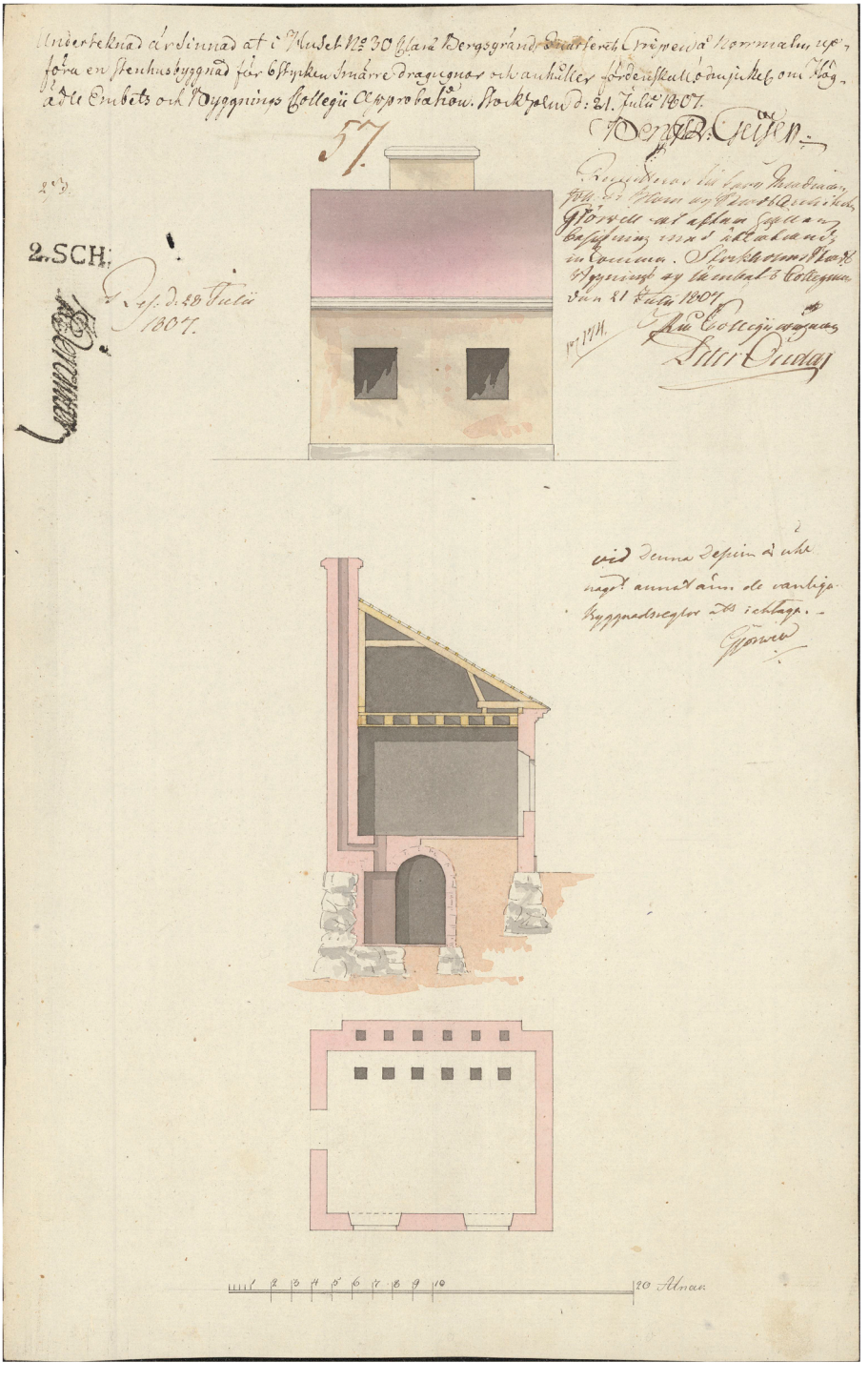


Figure 3. Building permit drawing for Gustaf Broling's steel works, by Carl Christoffer Gjörwell, 1807. Source: Stockholms stadsarkiv, Byggnadsnämndens expedition och stadsarkitektkontor, Bygglovsritningar 1713–1874, NS:37:1807:57. Erecting his steelmaking facilities in the quarter Gripen on Norrmalm, Broling took over where Bengt Qvist had left off, experimenting with the making of crucible steel. Reproduced with permission from Stockholms stadsarkiv.

a reform of technical education. From being largely organised internally in *Bergskollegium*, it came to involve other institutions, much thanks to the efforts of actors in *Jernkontoret*. The association's own 'mechanical' section took shape during the late-eighteenth century, and a School of Mechanics was created in Stockholm in 1798 (however, without the association's involvement). The early 1800s came to see further intensified debates, notably so concerning the founding of a special 'Mining School' (*Bergsskola*).[123]

From this reasoning, it might seem convenient to conclude that the influence over production, as well as the opportunities to participate in the circulation of useful knowledge, were gradually moved away from petty craftsmen. However, we must remember that a significant part of urban manufacturing was still carried out in small-scale businesses relying on manual dexterity and shop-floor training. Consequently, innovative projects and transfers of technical know-how were not limited to an elite group of academically educated experts, but rather involved actors from all levels of the early modern hierarchy of production. Artisans, like Carl Fredrik Ponsback, were involved not only in developing new business specialisations, but also in shaping the urban potentials for skill formation, codification, and the making of 'open technique'.

Ponsback was born and raised in a family of Stockholm cutlers. In 1790, being in his early twenties, he was still likely trained – together with his brothers and one additional apprentice – by his father Johan Fredrik in the family workshop on Norrmalm. Soon afterwards, he left Sweden to practise his craft abroad, acquiring 'the most essential knowledge' related to the making of surgical gear, cutting tools, scissors, razors, and fine knives. When advertising his services in 1805, he also emphasised 'the approbation of the most distinguished Doctors and Surgeons', a rather appropriate qualification for an aspiring instrument maker.[124] Five years later, Ponsback had settled with his family at the corner of the streets Styckgjutaregränden and Smedjegatan on Norrmalm, in a buzzling area with numerous small-scale metalworking businesses. Ponsback's workshop was no exception in this regard, as he only employed three apprentices, one of whom was his younger brother Christian Johan. Their production was increasingly geared towards surgical equipment, a sector that also employed a handful of other workplaces in the capital, including the one led by Gustaf Broling. Notwithstanding this competition, Ponsback was contracted by the Royal College of Medicine (*Collegium Medicum*) to make precision tools for army surgeons.[125] As Sweden was still involved in armed conflicts with its Baltic neighbours, this was certainly an important source of income. Still, Ponsback also targeted a wider group of consumers, promoting his skills in making 'all sorts of mechanical bandages, such as hernia trusses, [and] machines for hunchbacks and those who have crooked sides and feets'.[126]

The combination of state service and medical retailing remained important to Ponsback during the following years. In 1817, he moved his business to the 'Medical-Surgical Institute' on Kungsholmen, later known as the Karolinska Institute. The new workshop was housed in an orangery on the Institute's premises, and Ponsback was given responsibility for the instrument collections and for teaching students how to maintain the tools in daily surgical practice. In these tasks, he was likely assisted by his brother Christian J., who still worked by his side. The brothers complemented these duties with the training of instrument-making apprentices and the production of steel wares for a larger group of customers; knives, scissors, razors, and medical devices – made according to 'inventive ideas and models' – were advertised in the Stockholm newspapers and sold in a central retail shop as well as by order.[127]

By his death, in 1834, Ponsback had also trained a suitable successor for the job at the Karolinska Institute, Johan Albrecht Stille.[128] When founding his own company in the 1840s, the long-lasting supplier of surgical gear and medical tools Stille AB, he became one of the individuals who laid the foundations for a novel wave of economic and industrial growth in Stockholm. He was in the good company of entrepreneurs such as Jacques Lamm, Johan Wilhelm Bergström, and the brothers Jean and Carl Gerhard Bolinder, who founded their respective metalworking factories in the capital during the same decade. They were, in Ingrid Hammarström's words, men 'equipped with a thorough technical education', often complemented by study tours in England, who carried 'an unrestricted enthusiasm for the new era of mechanical engineering'.[129]

These business ventures not only pointed towards something new, however, as they, in many cases, were born out of a highly dynamic landscape of small-scale production, one that also became an important driving force behind the expansion for metal industries in Stockholm.[130] In this regard, the discussion above has pointed to an intricate mix of transnational movements, technical originality, and commercial strategies that gradually brought change about in the urban metalworking sector. When analysing such processes, we need to incorporate the undertakings of skilled people from the shop-floor world, like the Ponsback brothers. In the same way as noted by Hilaire-Pérez, they certainly played a key part in reshaping the structure of artisanal work, notably so by contributing to an intensified circulation of skills and technical know-how between domains of material production and education, as well as to the making of 'open technique' and enhanced potentials for the codification of useful craft methods. Moreover, and not to be forgotten, they bridged the spheres of manufacturing, trade, and consumption in new ways by a clever use of urban information networks in targeting growing markets.[131] Thus, they can definitely be said to have embodied the creative quest for improvement that characterised the capital's metal trades.

Such insights must of course be balanced with a thorough understanding of the 1760s crisis and its effects. However, also this part of the story is preferably built around the various individuals and groups who, in different ways, were active in the urban trades, some of whom we have encountered in this section. This analysis can help us see the diversity and nuances of pre-industrial production, but also the perceived ways ahead. In 1801, Johan Elers described both the expansive period during the first half of the 1700s and the more troublesome period after the crisis years. Some commentators counted the year of 1765 as a start for the 'waning and decay' of Stockholm manufacturing, but Elers also stressed that several (smaller and larger) workplaces had survived the tough decades. Moreover, promising new enterprises – Charles Apelquist's *fabrique* was brought up as one example – proved 'the benefit [that] Sciences have brought to the Manufactures'.[132] Regarding the capital's guilds, Elers indeed noted a negative trend, with declining numbers of masters, privilege-related conflicts, and a growing rivalry from rural handicraft production. Still, he also emphasised that the capital remained a vibrant centre for consumption and trade, and that widespread liberty, especially when it came to the making of 'coarser goods', could well be combined with strong town-based crafts. The path forward seemed obvious to Elers: '[i]t is not through reduced competition, but rather through extended industry that urban trades gain growth and the individual practitioner should seek his advantage'.[133]

Conclusion

Examining the impact of geographic mobility and the circulation of skills and technical know-how in the Stockholm metal trades before, during, and after the economic crisis of the 1760s, this article has thrown new light upon a period in the history of Stockholm that has recurrently been dealt with in economic-historical studies, often in terms of a long stagnation.[134] My ambition has been to move beyond this standard story, by highlighting three elements that have prominently featured in European research during the past two decades, although without comprehensive studies dealing with or including Scandinavian contexts. As a consequence, this article has provided important insights not only into developments in the Swedish capital but also into manufacturing, knowledge-making, and innovation in early modern European cities more generally.

First, this case adds important evidence regarding the transnational circulation of technical know-how and craft techniques in the European metal trades, emphasising the diverse movements, the variety of social interactions, and the assorted entanglements among travellers, local intermediaries, and outreaching administrative bodies involved in these processes.[135] The first empirical section demonstrated how the expansion of urban metalworking in Stockholm during the mid-eighteenth century depended on different forms of cross-border travelling, including traditional forms of tramping, state-sponsored study tours, and recruitments of foreign craftsmen or expatriate Swedes. These movements took place alongside the journeys made by other actors in the metal trades, like civil servants, and often incorporated a range of cities and manufacturing communities across Europe.[136] Expanding on such observations, I related these itineraries to local and regional forms of mobility, like the circulation of trainees and the relocations of state-recognised artisans, thereby adding more layers to the understanding of how the mutable urban context of manufacturing was interwoven with a likewise intricate transnational system of 'espionage' and 'intelligent travel'.[137]

This dynamic relationship is further illustrated by the crisis years, when metalworking artisans – many of whom were not native to Stockholm or had travelled across borders before – left the capital on a permanent or temporary basis, putting focus on the ways in which movements were actively used to navigate the economic and political fluctuations of a pre-industrial economy. The second empirical section also pointed to the fact that expanding metal industries in Britain became increasingly important after the crisis. British metalworking was indeed of significant interest earlier, to both state-funded mobile artisans and travelling civil servants such as Samuel Schröder, but the progress made in key processes (like casting and steelmaking) and the improved use of innovations such as the steam engine evidently affected the patterns of mobility through which technical know-how in the metal trades was transnationally circulated.[138]

Secondly, the discussion has taken new steps in highlighting an assortment of urban manufacturing practices, which jointly put focus on the circulation of knowledge that underpinned creativity, innovation, and skill formation in early modern towns, but which also played key parts in shaping the implementation of – and changes to – state-designed privilege systems and the making of administrative 'expertise'.[139] In scrutinising the expansion for metal manufacturing up until the mid-1760s, I discussed how the 'politicization of technology' and the making of 'open technique', with reference to Hilaire-Pérez, incorporated various forms of training, collaborative projects devoted to material testing, imitation, and product innovation, as well as attempts with new

organisational solutions.[140] These activities all help to explain how technical know-how from the outside was used, modified, debated, and further disseminated. There was indeed room for conflict in this changing context of production, often concerning production rights in the borderland between guilds and state-privileged manufacturing, but the discussion also emphasised how innovative accomplishments in the realm of everyday work gave rise to valuable interactions among the spheres of material production, trading, state supervision, and science. In this respect, the Stockholm case offers important novel perspectives to discussions on innovation and commercialisation in the eighteenth-century metal sector, which have largely been centred on Britain.[141]

Manufacturers who stayed in the capital during and after the turbulent crisis years were certainly faced with a more uncertain existence, in the light of cancelled state subsidies and the growing competition from Eskilstuna *fristad*. Still, the period from the late-1760s up until the turn of the century also saw a number of new enterprises being founded in the capital, increasingly inspired by British examples; English-style foundries, crucible steel works, and new *fabriques* for the improved (in some cases semi-mechanised) production of metal wares were created in a weakened, but prevailing, privilege-economy. In this regard, the 'expertise' of entrepreneurial administrators such as Bengt Qvist, Charles Apelquist, and Gustaf Broling are indeed signs of a move towards more industrial forms of metalworking.[142] However, I also stressed the need to recognise innovative initiatives that emerged out of a context of traditional – small-scale and often family-based – artisanal work, and which played an equally important part in promoting an 'extended industry', to use Elers' terms.

Third, and finally, the empirical sections in this article have strived to particularly accentuate the roles played by skilled craftspeople in the circulation of knowledge, illuminating how they performed various intermediary services in transfers of metalworking methods, but also actively engaged in creating new skills and novel forms of technical and organisational know-how, in line with Garrioch's reasoning.[143] Thus, and related to recent studies, they made key contributions both to the negotiation of useful knowledge and production rights in a context of cameralist state regulation, and to wider transnational processes by which craft techniques were codified.[144] The Swedish administrative archives lend themselves perfectly for further investigation on this topic, and the present study has consequently shown one possible way forward in a field of historical research that has so far lacked contributions regarding the northern periphery of an expanding European 'economy of codification'.[145]

With reference to Swedish economic-historical studies, it can be said that manufacturing in Stockholm saw significant improvements during the second half of the nineteenth century, following a long period of economic stagnation. The new wave of expansion can, somewhat simplistically, be dated to the liberal reforms of 1846 and 1864, marking a end for the old privilege order, and it was driven by 'capital- and skill-intensive sectors'.[146] Metalworking, 'in increasingly more industrial forms', became a leading sector (in terms of job creation) in the Stockholm economy, and it was characterised by both 'high quality production' and a higher average value of output (per worker) than other industries in the Swedish capital.[147] My intention has not been to question this story, but rather to emphasise the need for new perspectives when it comes to earlier phases of industrial development. I have shown that the period from the 1750s to the 1820s should by no means be categorically regarded as a downturn for the capital's trades. The movements and undertakings of state

officials, privileged manufacturers, and petty craftsmen in the metalworking sector rather illustrate a capacity for innovation, market adaptation, and renewal, observations that provide a significant complement to the idea of a 'stagnating metropolis'. Extending such a discussion one step further, my findings point to the need to not seek the move towards an industrial society in any sudden burst out of a static past, but instead aiming to 'qualify' this process through 'spatially aware' approaches that link together wider processes of circulation, the local dynamics of manufacturing and knowledge-making, and the various practices of skilled working people.[148]

Notes

1. RAS, Frihetstidens utskottshandlingar, R. 3494, Samuel Schröder, report ("Berättelse om Jern- Stål- och Metall-Fabrikerne i Riket för Åren 1765–1768"), April 4, 1769, 678, 716–20. All quotes from Swedish sources are, unless stated otherwise, my translations. On the impact of the economic crisis on the Swedish metal trades, see also Rydén, "Eskilstuna *Fristad*"; and Jansson, *Making Metal Making*, 77–82, 108–12.
2. Heckscher, *Sveriges ekonomiska historia från Gustav Vasa* 2:2, 598–621, 640–2, 757–81; and Nyström, *Stadsindustriens arbetare före 1800-talet*, 141–8, 251–7, 309–13.
3. e.g., Nyberg, ed., *Till salu*; Jakobsson, "Economic behaviour and social strategies in the Stockholm silk weaving industry, 1744–1831"; and Jansson, "Global Materials, Contact Zones, and Imitation."
4. Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*, 31–45, 209; Nyberg, *Kommersiell kompetens och industrialisering*; and Hammarström, *Stockholm i svensk ekonomi 1850–1914*. There are indeed more nuanced perspectives on developments in the Stockholm textile trades; see, e.g., Jakobsson, "Economic behaviour and social strategies in the Stockholm silk weaving industry, 1744–1831."
5. KB, Handskriftssamlingen, X. 283, Samuel Schröder, "Dagbok rörande Directeurs-Sysslan öfver Jern- Stål- och Metall-Fabrikerne i Riket af S Schröder," III, 1769, 18–21 (hereafter referred to as: Schröder, "Dagbok").
6. RAS, Frihetstidens utskottshandlingar, R. 3496, Thomas Lewis, memo, December 11, 1769, 332–3.
7. See, e.g. De Vito and Gerritsen, "Micro-Spatial Histories of Labour"; Hilaire-Pérez, "The Codification of Techniques"; Garrioch, "Introduction"; and Hilaire-Pérez and Lanoë, "Par-delà les pièges des discours."
8. The quoted term from Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*, see 8–30.
9. Evans and Rydén, *Baltic Iron*, 92–121; and Boëthius, *Magistraten och borgerskapet i Stockholm 1719–1815*, 92–6.
10. Schröder, "Dagbok," III, 1767, 43–50. On French manufacturing policies, see Hilaire-Pérez, "Invention and the State in 18th-Century France"; Minard, "Colbertism Continued?" Hilaire-Pérez, *L'invention technique au siècle des Lumières*, 51–82; and Belfanti, "Between Mercantilism and Market," especially 330–4.
11. Söderlund, *Stockholms hantverkarklass 1720–1772*, 60–70; Nyström, *Stadsindustriens arbetare före 1800-talet*, 188–219; Aldman, *En merkantilistisk början*, 141–53; and Jansson, *Making Metal Making*, 58–63, 98–104.
12. On the structure of Swedish textile manufacturing, see also Nyberg, "The Production of International Fashion."
13. Salvius, *Beskrifning öfver Sveriget*, 21, 36.
14. Berch, *Inledning til almänna hushålningen*, 219–25, 272–84.
15. Söderlund, *Stockholms hantverkarklass 1720–1772*, 117–49; Heckscher, *Sveriges ekonomiska historia från Gustav Vasa* 2:1, 524–34, 2:2, 599–601; and Jansson, *Making Metal Making*, 97–104.
16. SSA, Hall- och manufakturrätten, B3:1 and B3:5, Stockholm *Hallrätt*, reports, (*fabriksberättelser*), 1740 and 1760.

17. It must be pointed out that also the head tax records have their shortcomings, and cannot therefore be regarded as entirely complete sources; see Söderlund, *Stockholms hantverkarklass 1720–1772*, 117–22.
18. Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*, 31–64, 150–208 (quotations at pages 31, 202, and 206).
19. Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*, 34–6. To a large extent, Söderberg *et al* extend on previous studies by Eli F. Heckscher and Per Nyström; see note 2 above, and Heckscher "De svenska manufakturerna under 1700-talet."
20. Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*, 35, 39–42, 44 (for quotation).
21. See, e.g. Edgren, *Lärling – gesäll – mästare*, 76–87.
22. e.g., Ohlsson, *Eskilstuna fristad*; and Isacson and Magnusson, *Proto-industrialisation in Scandinavia*, 51–108.
23. On spatial changes with regard to craftwork and manufacturing in Stockholm after the economic crisis, see also Hayen, *Stadens puls*, 150–9.
24. Roche, *A History of Everyday Things*, 36; and De Munck and Romano, "Knowledge and the Early Modern City," 3.
25. Clemente, "Innovation in the Capital City," 322. One could stretch this discussion further back in time, and use other concepts to describe encounters between different groups and forms of knowledge; see, e.g. Pamela Long's discussion of the city as a "trading zone": Long, *Artisan/Practitioners*, 112–26.
26. Epstein, "Labour Mobility"; and Reith, "Circulation of Skilled Labour." See also Prak, Haru Crowston, De Munck, Kissane, Minns, Schalk, and Wallis, "Access to the Trade."
27. See, e.g. Bartlett, *Human Capital*, 143–79; Harris, *Industrial Espionage and Technology Transfer*, 174–204; and Belfanti, "Between Mercantilism and the Market."
28. Davids and De Munck, "Innovation and Creativity," quote from 27; Hilaire-Pérez and Verna, "Dissemination of Technical Knowledge"; and Garrioch, "Introduction," 3–9, 17–25.
29. Lilja, *Tjuvehål och stolta städer*, 116–22
30. Söderlund, *Stockholms hantverkarklass 1720–1772*, 253–7; and Candréus, "Dealing with Mobility."
31. Amelin, *Medaljens baksida*; Hinners, *De fransöske handtwerkarne*, especially 83–151; and Jansson, *Making Metal Making*, 173–210.
32. Christensen, *Det moderne projekt*, 267–76, 435–44; and Gustafsson, "Rörlig kunskap och flyttbara färdigheter."
33. Lindqvist, *Technology on Trial*, 97–9; and Fors, *Limits of Matter*, 46–52, 103–9.
34. Raj, *Relocating Modern Science*, 18–22, 225–6; Roberts, "The Circulation of Knowledge in Early Modern Europe"; and Smith, "Nodes of Convergence, Material Complexes, and Entangled Itineraries." I also take inspiration from recent studies on the Scandinavian metal trades; e.g. Hutchison, "Exploring an Early Cross-border Trade System"; Ranestad, "Copper Trade and Production of Copper, Brass and Bronze Goods in the Oldenburg Monarchy"; and Jansson and Rydén, "Improving Swedish Steelmaking."
35. See also De Munck and Soly, "'Learning on the Shop Floor' in Historical Perspective"; Wallis, "Apprenticeship and Training in Premodern England"; Davids and De Munck, "Innovation and Creativity," 20–2; and Wallis, "Between Apprenticeship and Skill."
36. Sonenscher, *Work and Wages*, 22–9, 130–73; and Riello, "Strategies and Boundaries." Similar arguments have been made regarding the late-eighteenth-century Stockholm textile trades; see Nyberg, *Köpes: ull, säljes: kläde*.
37. Berg, *The Age of Manufactures*, 255–79; Clifford, "Concepts of Invention, Identity and Imitation"; Styles, "Product Innovation in Early Modern London"; Berg, "From Imitation to Invention"; Berg, *Luxury and Pleasure*, 85–110, 154–92; and Evans and Withey, "An Enlightenment in Steel?" On innovation in early modern societies, see also MacLeod, *Inventing the Industrial Revolution*, 40–57, 75–143; Hilaire-Pérez, *L'invention technique au siècle des Lumières*, above all 51–142; and Lucas "Manufacturing Modernity."

38. The concept of the bazaar was introduced by Michael Sonenscher in his study of the eighteenth-century French trades; see Sonenscher, *Work and Wages*, 22–9 (quotation at page 29), 135–40. In the 2012 new edition of the book, however, he highlighted the existence of "a better term," used by eighteenth-century commentators, namely "fashion's empire" (*L'empire de la mode*); see Sonenscher, *Work and Wages*, xii–xiii.
39. De Munck and Romano, "Knowledge and the Early Modern City," 18.
40. e.g., Belhoste, "Les hommes du fer"; Luu, *Immigrants and the Industries of London, 1500–1700*, 219–58. For a Swedish context: Douhan, *Arbete, kapital och migration*. On "mobile artisans" and the complex early modern world of craftwork: Garrioch, "Introduction," 3–9.
41. Schaffer, Roberts, Raj, and Delbourgo, "Introduction," xiv, xxii.
42. Garrioch, "Introduction," 16. The emphasis on pre-industrial artisans as active knowledge-makers is indeed not new. For an earlier, and highly influential example, see Smith, *The Body of the Artisan*.
43. Hilaire-Pérez, "Technology as a Public Culture"; Bertucci, *Artisanal Enlightenment*, 17–23, 79–175; and Hilaire-Pérez, "The Codification of Techniques." For nuanced perspectives on early modern practical knowledge; see also Roberts and Schaffer, "Preface"; Valleriani, "The Epistemology of Practical Knowledge," 12–14; Garrioch, "Introduction," 10–25; and Jones, "In Search of the 'Invisible' Artisan."
44. See Hilaire-Pérez and Lanoë, "Par-delà les pièges des discours," 362–3, 373–4, 389–90. One could also speak of the "technological life" of artisans; see Alder, *Engineering the Revolution*, 175–84 (and, more generally, xii, 153, 229).
45. The quote from De Vito and Gerritsen, "Micro-Spatial Histories of Labour," 11. Recent studies have not only nuanced predominant modern perspectives on creativity and urban knowledge formation (see, e.g. Florida, *The Rise of the Creative Class*), but also what is perceived as uncritical reproductions of early modern representations of artisanal work, mostly stemming from political and scientific elites; see De Munck, "Artisans as Knowledge Workers"; De Munck, "Shared Skills and Technologies of Community Formation"; and Hilaire-Pérez and Lanoë, "Par-delà les pièges des discours."
46. Rydberg, *Svenska studieresor till England under Frihetstiden*, 139–201; and Lindqvist, *Technology on Trial*. For recent studies that offer more nuanced perspectives, see Rydén, "The Enlightenment in Practice"; Rydén, "Eskilstuna *Fristad*"; Fors, *Limits of Matter*; and Fors and Orrje, "Describing the World and Shaping the Self." On "industrial espionage" and "intelligent travel," see Bertucci "Enlightened Secrets."
47. Jansson, "Global Materials, Contact Zones, and Imitation"; Orrje, "The Logistics of the Republic of Letters"; and Almbjär and Clemente, "Mercantilism from Below?"
48. Hilaire-Pérez, "Technology as a Public Culture," 139–43; Pérez, "Inventing in a World of Guilds," 236–53; and Hilaire-Pérez, "The Codification of Techniques," 448.
49. Schröder, "Dagbok," I, 1756, 74–6; RAS, Kommerskollegium, huvudarkivet, BIa:137, *Kommerskollegium*, registry protocol, June 2, 1755; EVIc:26, A. Wynantz, London, 1748–56, letter from Arnold Wynantz to *Kommerskollegium*, July 4, 1755; RAS, Diplomatica Anglica, vol. 366, various letters from *Kommerskollegium* and *Krigskollegium* to Arnold Wynantz, 1755 and 1756. See also Jansson, *Making Metal Making*, 159–60. On merchants Spalding and Brander, see Orrje, "The Logistics of the Republic of Letters."
50. Amelin, *Medaljens baksida*, 52–6, 163; and Orrje, "Patriotic and Cosmopolitan Patchworks."
51. Schröder, "Dagbok," I, 1756, 135–6, 144, and II, 1757, 89–90; and Schröderstierna, *Berättelser*, II, 65.
52. RAS, Frihetstidens utskottshandlingar, R. 2766, no. 42, Matthias Holmerus, memo, January 3, 1741; RAS, Frihetstidens utskottshandlingar, R. 2899, Matthias Holmerus, memo, January 16, 1747, 509–13; Westrin, *Nordisk familjebok*, vol. 18, 886; and Wollin, *Svenska stilgjuteriet*, 9–11.

53. RAS, Frihetstidens utskottshandlingar, R. 3600, Anders Dahlström, report and list of expenditures, 1772, 255–6; and Schröder, "Dagbok," I, 1756, 162–3.
54. On the privileges of Swedish manufacturing artisans: Söderlund, *Stockholms hantverkarklass 1720–1772*, 298; Nyström, *Stadsindustriens arbetare före 1800-talet*, 264–6. The links between artisanal mobility and special privileges have also been emphasised for a wider European context; see, e.g. Belfanti, "Between Mercantilism and Market."
55. SSA, Hall- och manufakturrätten, B3:1–2, Stockholm *Hallrätt*, reports, 1742–47; SSA, Överståthållarämbetet för uppbördsärenden, Kamrerareexpeditionen, G1AA:53, tax register (*kronotaxeringslängd*), Stockholm, 1750, 201; and Wikström, "Peter Momma."
56. SSA, Hall- och manufakturrätten, B3:4–7, Stockholm *Hallrätt*, reports, 1757–66; SSA, Överståthållarämbetet för uppbördsärenden, Kamrerareexpeditionen, G1BA:21:3, head tax record (*mantalslängd*), Stockholm, 1760, Klara västra övre och yttre och Kungsholmen, 209; and Schröderstierna, *Berättelser*, II, 69–70.
57. *Manufakturkontoret*, protocols, February 11 and April 27, 1754, quoted in Ronnestam, *Fredmans tid*, 246.
58. RAS, Frihetstidens utskottshandlingar, R. 3598, Augustin Bourdillon, memo, August 29, 1771, 1098; Ronnestam, *Fredmans tid*, 250–60.
59. Schröder, "Dagbok," I, 1756, 83–5; Schröderstierna, *Berättelser*, II, 68–9, 70–1; RAS, Frihetstidens utskottshandlingar, R. 3600, Dahlström, list of expenditures, 256; and Lagerquist, "Argent haché," 11.
60. Schröderstierna, *Berättelser*, I, 46–7, and II, 51.
61. KB, Handskriftssamlingen, X. 303, Samuel Schröder, "Dagbok rörande Handel, Näringar och Manufakturer m.m. Uti Danmark, Holland, England, Frankrike och Tyskland. Under verkställde resor, åren 1748–1751 förd af Samuel Schröder," IV, 1026–7.
62. SSA, Hall- och manufakturrätten, E3:3, *Kommerskollegium*, privilege for Hartwig Losch, December 9, 1755, 194–7; and Schröderstierna, *Berättelser*, II, 67–8.
63. Schröder, "Dagbok," I, 1754, 180–2.
64. Schröder, "Dagbok," II, 1756, 9–14; RAS, Manufakturkontoret, huvudarkivet, D4:172, no. 10, Anders Lissander, Johan F. Kryger, Johan A. Ernst, and Jakob J. Westberg, memo, December 15, 1756.
65. In demographic discussions, this tendency has been referred to as an "urban graveyard effect." As shown by Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*, 150–200, the mortality in Stockholm was exceptionally high during the period of interest here, compared to other European cities. Still, one must also relate such trends to varieties of migration and knowledge dissemination to fully understand the dynamics of pre-industrial urban economies; see Davids and De Munck, "Innovation and Creativity," 22–7.
66. See also Jansson, *Making Metal Making*; and Jansson, "Global Materials, Contact Zones, and Imitation."
67. On this subject, see Pérez, "Inventing in a World of Guilds," especially 245–53; Pérez, "Technology, Curiosity and Utility"; Wallis and Wright, "Evidence, Artisan Experience, and Authority"; and Ashworth, "Quality and the Roots of Manufacturing 'Expertise' in Eighteenth-Century Britain."
68. RAS, Frihetstidens utskottshandlingar, R. 2992, no. 69, Jacob Faggot, memo, May 12, 1752, and *Manufakturkontoret*, memo, May 6, 1752; and Ronnestam, *Fredmans tid*, 278–87.
69. Schröder, "Dagbok," II, 1758, 2–5; RAS, Frihetstidens utskottshandlingar, R. 3495, Friedrich Heuman, Petter Ernst, Eric Lindgren *et al.*, memo, undated (most likely written in 1765), 616–17.
70. Schröder, "Dagbok," II, 1758, 5–7, 73, and 1759, 12–13; RAS, Manufakturkontoret, Kamrerarekontoret, C3:346, verification of production, Olof Beckelin, June 19, 1760, 912, and C3:349, verification of production, Olof Beckelin, April 23, 1761, 592. See also Ronnestam, *Fredmans tid*, 110–14.
71. See also Jansson, *Making Metal Making*, 189–92.
72. Schröderstierna, *Berättelser*, II, 65–6; RAS, Manufakturkontoret, Kamrerarekontoret, C3:349, verifications of apprentice training, Jacob C. Ketscher, March 17 and

November 18, 1761, 367, 1577; Schröder, "Dagbok," II, 1758, 23–27; 1759, 1–2. See also Jansson, *Making Metal Making*, 189–92, 202–5.

73. RAS, Frihetstidens utskottshandlingar, R. 3600, Dahlström, report, 1772, 255; and Schröderstierna, *Berättelser*, II, 59, 69–70.
74. Schröder, "Dagbok," II, 1757, 70–2; SSA, Hall- och manufakturrätten, B3:5–7, Stockholm *Hallrätt*, reports, 1760–66.
75. SSA, Hall- och manufakturrätten, B3:7, Stockholm *Hallrätt*, reports, 1765–66; RAS, Frihetstidens utskottshandlingar, R. 3598, Bourdillon, memo, August 29, 1771, 1099–1100.
76. On craft training outside the guilds, see Davids, "Apprenticeship and Guild Control"; Caracausi, "Textiles Manufacturing"; and Wallis, "Between Apprenticeship and Skill." In Britain, education in technical drawing for artisans were realised mainly through private initiatives; see Puetz, "Design Instruction for Artisans in Eighteenth-Century Britain."
77. Schröderstierna, *Berättelser*, II, 80–2; RAS, Frihetstidens utskottshandlingar, R. 3494, Schröder, report, April 4, 1769, 697–9.
78. Clifford, "Concepts of Invention, Identity and Imitation"; Berg, *Luxury and Pleasure*, 158–79; Pérez, "Technology, Curiosity and Utility"; Evans and Withey, "An Enlightenment in Steel?"; and Hilaire-Pérez, "The Codification of Techniques," especially 451–6.
79. Schröder, "Dagbok," II, 1757, 8–10, 69–70; 1758, 27–8; and Schröderstierna, *Berättelser*, II, 66–7.
80. Schröderstierna, *Berättelser*, II, 70; *Dagligt allehanda*, July 21, 1769. Accessed April 22, 2024, via https://tidningar.kb.se. See also Jansson, "Global Materials, Contact Zones, and Imitation," 55–61.
81. For examples, see Schröder, "Dagbok," I, 1756, 145–6; 1757, 13–17. For an elaborated discussion on the making and testing of steel during the mid-eighteenth century, see Jansson and Rydén, "Improving Swedish Steelmaking."
82. The quotes from Smith, "Making as Knowing," 40.
83. Schröder, "Dagbok," II, 1757, 227–9.
84. Schröder, "Dagbok," II, 1759, 54–6; 1760–1, 35–40.
85. RAS, Manufakturkontoret, huvudarkivet, D5:180, Petter Ernst, report on Verdigris ("Företal. Angående Spansk Grönas tillwärckande och beredning, så wäl af den gröfre, som finare Sorten distilerade och rafinerade färgen"), October 26, 1761; Lagerquist, "Petter Ernst."
86. The quote from Pérez, "Inventing in a World of Guilds," 253.
87. RAS, Kollegiers m fl, landshövdingars, hovrätters och konsistoriers skrivelser till Kungl Maj:t, 9:154, *Kommerskollegium*, statement, November 4, 1768.
88. Alder, *Engineering the Revolution*, above all 127–62, 221–49. For process inventions in the British metal trades, see also Berg, *The Age of Manufactures*, 266–71.
89. See also Jansson, *Making Metal Making*, 127–31; and Jansson, "Global Materials, Contact Zones, and Imitation."
90. Schröderstierna, *Berättelser*, II, 35–7; Schröder, "Dagbok," I, 1753, 42–3, 1756, 55–6; RAS, Frihetstidens utskottshandlingar, R. 3495, Petter Ernst and Johan Lindquist, memo, October 1769, 567–8. See also Ronnestam, *Fredmans tid*, 105–14.
91. RAS, Frihetstidens utskottshandlingar, R. 3495, *Handels- och manufakturdepuationen*, report, September 1, 1766, 869; R. 3494, Schröder, report, April 4, 1769, 683–4.
92. De Vito and Gerritsen, "Micro-Spatial Histories of Labour," 15. Similar arguments are made, e.g. by Roberts and Schaffer, "Preface," XX – XXI; and Garrioch, "Introduction," 17–25.
93. The quoted term from Ashworth, "Quality and the Roots of Manufacturing 'Expertise' in Eighteenth-Century Britain." See also Alder, *Engineering the Revolution*, 56–124. On the much-debated concepts of "experts" and "expertise," see, e.g. Ash, "By Any Other Name."
94. RAU, Eskilstuna rådhusrätt och magistrat, F13:7, Samuel Schröder, report ("Berättelse om Jern- Stål- ock Metall-Fabrikerne i Riket. Aflämnad til Riksdagen År 1765"), December, 1764, 141, 144–8; RAS, Frihetstidens utskottshandlingar, R. 3494, Schröder, report, April 4, 1769, 712–13; Ronnestam, *Fredmans tid*, 257–60; and Amelin, *Medaljens baksida*, 114–16.

95. RAU, Eskilstuna rådhusrätt och magistrat, F13:7, Schröder, report, December, 1764, 143; RAS, Frihetstidens utskottshandlingar, R. 3496, copy of letter, Carl Ribbing to Claes Ekeblad Jr., Saint Petersburg, July 21, 1769, 406–7. See also Bartlett, *Human Capital*, 162–3.
96. RAS, Frihetstidens utskottshandlingar, R. 3494, Schröder, report, April 4, 1769, 717–18.
97. Schröder, "Dagbok," III, 1765, 5–8, 1766, 57–8, 1767, 7–11, 16; RAS, Frihetstidens utskottshandlingar, R. 3335, *Manufakturkontoret*, memo, December 19, 1765, 59–60; RAS, Kollegiers m fl … skrivelser till Kungl Maj:t, 11:3:90, *Kommerskollegium* and *Krigskollegium*, statement, February 9, 1780. Thanks to Fiona Ryder, who kindly made me aware of the latter source. Thanks also to Bryan and Robin Carsberg for exchanging information about Georg(e) N. Carsberg and the Carsberg family business.
98. Osborn, *Essays on the practice of midwifery*, 96.
99. Hayen and Nyberg, "Konkursinstitutet i Stockholm 1720–1850," 37–9.
100. RAS, Frihetstidens utskottshandlingar, R. 3335, *Manufakturkontoret*, memo, April 17, 1766, 321–3.
101. Schröder, "Dagbok," III, 1764, 15–22, 1765, 4–5, 1768, 10–13, 25–6, 49–50; *Stockholms Posttidningar*, January 17, 1771, 4. Accessed April 22, 2024 via https://tidningar.kb.se; and SSA, Hall- och manufakturrätten, B3:9–11, Stockholm *Hallrätt*, reports, 1772–78.
102. RAS, Frihetstidens utskottshandlingar, R. 3494, Schröder, report, April 4, 1769, 703–8.
103. See Melkersson, *Staten, ordningen och friheten*, 161–73; Jansson, *Making Metal Making*, 77–82; and Jansson and Rydén, "Making and Trading Metals." Similar points have been raised in discussions of Colbertism and liberalism in the context of eighteenth-century French manufacturing; see, e.g. Minard, "Colbertism Continued?"
104. RAS, Frihetstidens utskottshandlingar, R. 3600, *Kommerskollegium*, memo, March 4, 1772, 232–4, and Samuel Schröder, memo, April 13, 1772, 237–9; R. 3599, Petter Åberg, memo, August 5, 1771, 36–37, and *Kommerskollegium*, memo, September 28, 1771, 39–41.
105. RAS, Frihetstidens utskottshandlingar, R. 3494, Schröder, report, April 4, 1769, 721–3.
106. RAS, Frihetstidens utskottshandlingar, R. 3496, Samuel Schröder and Samuel Sandels, statement, December 12, 1769, 334–5; Boëthius, *Jernkontorets historia. D. 3:1*, 81; and Kjellander, "Thomas Lewis."
107. *Stockholms Posttidningar*, September 16, 1771, 4. Accessed April 22, 2024 via https://tidningar.kb.se.
108. Boëthius, *Jernkontorets historia. D. 3:1*, 81; SSA, Stockholms magistrat och rådhusrätt, F6a:63, 8/1774, bankruptcy act, Thomas Lewis, May 5, 1773.
109. RAS, Frihetstidens utskottshandlingar, R. 3599, Thomas Lewis, memo, September 18, 1771, 89–94; and Kjellander, "Thomas Lewis."
110. KTHB, Kungl. Tekniska högskolans biblioteks manuskriptsamling, Bergsskolans biblioteks manuskript, D7:2a, undated memo ("Memorandum concerning Mess:rs Boulton and Watts new invented Steam Engines"); D7:2b, letter from Charles Gascoigne to Thomas Lewis, Carron, April 22, 1775; D7:2c, undated memo ("Om Gjuteriernas för stora tyngd").
111. Simonsson, "Daniel Asplund." The company was split in two parts, Bergsund and Nyhyttan, and Asplund only owned the former. In 1812, the two workplaces were reunited under one owner, Gustaf Daniel Wilcke; see Boëthius, *Jernkontorets historia. D. 3:1*, 82.
112. Evans and Rydén, *Baltic Iron*, 250–91; and Jansson and Rydén, "Rörelser mot det industriella."
113. For similar perspectives, see Hilaire-Pérez, "Invention and the State in 18th-Century France"; Alder, *Engineering the Revolution*, especially 16–19, 344–51; Hilaire-Pérez, *L'invention technique au siècle des Lumières*, 51–82; Ashworth, "Quality and the Roots of Manufacturing 'Expertise' in Eighteenth-Century Britain"; Hilaire-Pérez, "The Codification of Techniques," above all 447–51. On the promises of studying state-making from an epistemological angle; see also Felten and von Oertzen "Bureaucracy as Knowledge."
114. KTHB, Bergsskolans biblioteks manuskript, E12:5, Bengt Qvist Andersson, report ("Beskrifning om Gjutståls beredningen"), December 15, 1769. See also Jansson and Rydén, "Improving Swedish Steelmaking," 532–3.

115. TMA, Sven Rinmans arkiv, F1K:2, Bengt Qvist Andersson, copy of memo, April 26, 1787, with attached statements: Eric Lindgren, April 28, 1784, and Jean-Jacques Perret, autumn of 1783. On Qvist's steel works, see also Boëthius, *Jernkontorets historia. D. 3:1*, 76–7.
116. TMA, Sven Rinmans arkiv, F1K:2, Bengt Qvist Andersson, copy of memo, April 26, 1787; Boëthius, *Jernkontorets historia. D. 3:1*, 78–9.
117. RAS, Kollegiers m fl . . . skrivelser till Kungl Maj:t, 9:201, Charles Apelquist, application, 1777; 9:234, *Kommerskollegium*, statement, March 7, 1787; Bergman, "Charles Apelquist," 130–2; Amelin, *Medaljens baksida*, 147–8; Petterson, "Samuel Owen," 460.
118. *Dagligt allehanda*, June 12, 1788. Accessed April 22, 2024 via https://tidningar.kb.se; SSA, Hall- och manufakturrätten, B3:15–16, Stockholm *Hallrätt*, reports, 1788 and 1789; RAS, Kollegiers m fl . . . skrivelser till Kungl Maj:t, 9:234, Bengt Qvist Andersson, statement, 1787, and Charles Apelquist and H. C. Holmström, application, 1787; and Bergman, "Charles Apelquist," 134–7.
119. Bergman, "Charles Apelquist," 138–43; and Petterson, "Samuel Owen."
120. RAS, Kollegiers m fl . . . skrivelser till Kungl Maj:t, 9:285, Gustaf Broling, application, 1801; *Kongl. Vetenskaps-Academiens handlingar för år 1838*, 302–8; Boëthius, *Jernkontorets historia. D. 3:1*, 356–65. In his printed travel report, Broling commented on having met and conversed with Huntsman during his stay in Attercliffe; see Broling, *Anteckningar*, D. 2, 148–9.
121. RAS, Kollegiers m fl . . . skrivelser till Kungl Maj:t, 9:285, Broling, application, 1801.
122. Jansson, "Nipper och nyttigt smide"; and Jansson and Rydén, "Making and Trading Metals."
123. The School of Mechanics was incorporated into the Technical Institute (later the KTH Royal Institute of Technology), in full operation from 1827. A Mining School was founded in 1819, and operated in Falun from 1822. See Boëthius, *Jernkontorets historia. D. 3:1*, 39–43, 124–41, 268–79; Torstendahl, *Teknologins nytta*, 18–26, 55–61.
124. SSA, Överståthållarämbetet för uppbördsärenden, Kamrerareexpeditionen, G1BA 25:9, head tax record, Stockholm, 1790, Klara övre och Kungsholmen, 83; and *Dagligt allehanda*, October 17, 1805. Accessed April 22, 2024 via https://tidningar.kb.se.
125. SSA, Överståthållarämbetet för uppbördsärenden, Kamrerareexpeditionen, G1BA 27:4, head tax record, Stockholm, 1810, Jakob, 331; *Dagligt allehanda*, October 19, 1811. Accessed 22 April, 2024 via https://tidningar.kb.se; RAS, Kollegiers m fl . . . skrivelser till Kungl Maj:t, 9:328, Carl Fredrik Ponsback, application, 1810, and *Collegium Medicum*, copy of protocol, November 11, 1811.
126. *Dagligt allehanda*, October 19, 1811.
127. Jülich, "Johan Albrecht (Albert) Stille"; *Dagligt allehanda*, April 28, 1817 and December 19, 1818. Accessed April 22, 2024 via https://tidningar.kb.se; and *Inrikes Tidningar*, April 28, 1818. Accessed April 22, 2024 via https://tidningar.kb.se.
128. Jülich, "Johan Albrecht (Albert) Stille"; and Halverson, *Tools of the Trade*, 34–40.
129. Hammarström, *Stockholm i svensk ekonomi 1850–1914*, 295–310 (quotation at page 297). See also Gårdlund, "Teknik och tekniker i den tidiga svenska verkstadsindustrien."
130. See Hammarström, Stockholm i svensk ekonomi 1850–1914, 309–10.
131. Hilaire-Pérez, "Technology as a Public Culture"; Pérez, "Inventing in a World of Guilds"; Hilaire-Pérez and Rabier, "Self-Machinery?" above all 20–1; and Hilaire-Pérez, "The Codification of Techniques."
132. Elers, *Stockholm. III. Delen*, 354–7.
133. Elers, *Stockholm. III. Delen*, 358–77 (longer quotation at page 377). On trade in the capital, see also Ibid., 290–332.
134. See, above all, Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*.
135. In these respects, I have taken inspiration from nuanced perspectives on both early modern artisanal work and the circulation of skills and technical know-how; see, especially, Hilaire-Pérez and Verna, "Dissemination of Technical Knowledge"; Schaffer, Roberts, Raj, and Delbourgo, "Introduction"; Roberts, "The Circulation of Knowledge in Early Modern Europe"; Garrioch, "Introduction."

136. See also Rydén, "The Enlightenment in Practice"; Fors and Orrje, "Describing the World and Shaping the Self".
137. Bertucci "Enlightened Secrets."
138. On this matter, see also Jansson and Rydén, "Rörelser mot det industriella."
139. Belfanti, "Between Mercantilism and Market"; Ashworth, "Quality and the Roots of Manufacturing 'Expertise' in Eighteenth-Century Britain"; Davids and De Munck, "Innovation and Creativity"; and De Munck and Romano, "Knowledge and the Early Modern City."
140. Hilaire-Pérez, "Technology as a Public Culture," 139–43; Pérez, "Inventing in a World of Guilds," 236–53; and Hilaire-Pérez, "The Codification of Techniques," 448.
141. See Clifford, "Concepts of Invention, Identity and Imitation"; Styles, "Product Innovation in Early Modern London"; Berg, "From Imitation to Invention"; Berg, *Luxury and Pleasure*, 85–110, 154–92; and Evans and Withey, "An Enlightenment in Steel?"
142. See also Jansson and Rydén, "Improving Swedish Steelmaking"; and Jansson and Rydén, "Rörelser mot det industriella."
143. Garrioch, "Introduction," 10–25.
144. See, e.g. Bertucci, *Artisanal Enlightenment*, 17–23, 79–175; and Hilaire-Pérez, "The Codification of Techniques."
145. The quoted term from Valleriani, "The Epistemology of Practical Knowledge," 12–14.
146. Söderberg, Jonsson, and Persson, *A Stagnating Metropolis*, 31–45, 209; Nyberg, *Kommersiell kompetens och industrialisering*; and Hammarström, *Stockholm i svensk ekonomi 1850–1914*.
147. Hammarström, *Stockholm i svensk ekonomi 1850–1914*, 321.
148. See De Munck, "Artisans as Knowledge Workers," 228–30; De Vito and Gerritsen, "Micro-Spatial Histories of Labour," 15; Hilaire-Pérez and Lanoë, "Par-delà les pièges des discours"; and Jones, "In Search of the 'Invisible' Artisan."

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Bibliography

Archives

Kungliga biblioteket (KB), Stockholm [The Royal Library]: Handskriftssamlingen.
Kungliga Tekniska högskolans bibliotek (KTHB), Stockholm [The library of The KTH Royal Institute

of Technology: Kungl. Tekniska högskolans biblioteks manuskriptsamling, Bergsskolans biblioteks manuskript.
Riksarkivet i Stockholm (RAS) [The National Archives of Sweden, Stockholm]: Diplomatica Anglica; Frihetstidens utskottshandlingar; Kollegiers m fl landshövdingars, hovrätters och konsistoriers skrivelser till Kungl. Maj:t; Kommerskollegium, huvudarkivet; Manufakturkontoret, huvudarkivet; Manufakturkontoret, Kamrerarekontoret.
Stockholms stadsarkiv (SSA), Stockholm [The City Archive of Stockholm]: Hall- och manufakturrätten; Stockholms magistrat och rådhusrätt; Överståthållarämbetet för uppbördsärenden, Kamrerareexpeditionen (online access: https://sok.riksarkivet.se/arkiv/ARNV8mrnFKAxKS6nK99qG0).
Tekniska museets arkiv (TMA), Stockholm [The Archive of the National Museum of Science and Technology]: Sven Rinmans arkiv.
Riksarkivet i Uppsala (RAU) [The National Archives of Sweden, Uppsala]: Eskilstuna rådhusrätt och magistrat.

Newspapers

Dagligt allehanda. https://tidningar.kb.se .
Inrikes Tidningar. https://tidningar.kb.se.
Stockholms Posttidningar. https://tidningar.kb.se.

References

Alder, K. *Engineering the Revolution: Arms and Enlightenment in France, 1763–1815*. Princeton, NJ: Princeton University Press, 1997.
Aldman, L.-A. *En merkantilistisk början: Stockholms textila import 1720–1738*. Uppsala: Acta Universitatis Upsalisensis, 2008.
Almbjär, M., and A. Clemente. "Mercantilism from Below? Swedish Consuls and Merchant Networks Between Naples and Stockholm in the Mid Eighteenth Century." *Business History* 67, no. 1 (2025): 105–125. doi:10.1080/00076791.2023.2227574.
Amelin, O. *Medaljens baksida: instrumentmakaren Daniel Ekström och hans efterföljare i 1700-talets Sverige*. Uppsala: Institutionen för idé- och lärdomshistoria, Uppsala University, 1999.
Ash, E. H. "By Any Other Name: Early Modern Expertise and the Problem of Anachronism." *History and Technology* 35, no. 1 (2019): 3–30. doi:10.1080/07341512.2019.1608082.
Ashworth, W. J. "Quality and the Roots of Manufacturing 'Expertise' in Eighteenth-Century Britain." *Osiris* 25, no. 1 (2010): 231–254. doi:10.1086/657271.
Bartlett, R. P. *Human Capital: The Settlement of Foreigners in Russia 1762–1804*. Cambridge: Cambridge University Press, 1979.
Belfanti, C. M. "Between Mercantilism and Market: Privileges for Invention in Early Modern Europe." *Journal of Institutional Economics* 2, no. 3 (2006): 319–338. doi:10.1017/S1744137406000439.
Belhoste, J.-F. "Les hommes du fer: Réflexions sur l'émergence et la diffusion de l'innovation (XIVe–XVIIe siècle)." In *Le technicien dans la cité en Europe occidentale, 1250–1650*, edited by M. Arnoux and P. Monnet, 271–289. Rome: École française de Rome, 2004.
Berch, A. *Inledning til almänna hushålningen, innefattande grunden til politie, oeconomie och cameral wetenskaperne: til deras tiänst, som biwista de almänne föreläsningar inrättad/af Anders Berch*. Stockholm: Salvius, 1747.
Berg, M. *The Age of Manufactures, 1700–1820: Industry, Innovation and Work in Britain*. 2nd ed. London: Routledge, 1994.
Berg, M. "From Imitation to Invention: Creating Commodities in Eighteenth-Century Britain." *The Economic History Review* 55, no. 1 (2002): 1–30. doi:10.1111/1468-0289.00212.
Berg, M. *Luxury and Pleasure in Eighteenth-Century Britain*. Oxford: Oxford University Press, 2005.
Bergman, G. "Charles Apelquist: En pionjär inom den mekaniska verkstadsindustrien i Sverige." *Dædalus* 1952 (1952): 129–146.

Bertucci, P. "Enlightened Secrets: Silk, Intelligent Travel, and Industrial Espionage in Eighteenth-Century France." *Technology and Culture* 54, no. 4 (2013): 820–852. doi:10.1353/tech.2013.0123.

Bertucci, P. *Artisanal Enlightenment: Science and the Mechanical Arts in Old Regime France*. New Haven: Yale University Press, 2017.

Boëthius, B. *Magistraten och borgerskapet i Stockholm 1719–1815*. Stockholm: Norstedt, 1943.

Boëthius, B. *Jernkontorets historia. D. 3:1. Jernkontoret och tekniken före götstålsprocesserna*. Stockholm: Norstedt, 1955.

Broling, G. *Anteckningar under en resa i England, åren 1797, 1798 och 1799, af Gustaf Broling på Bruks Societetens anmodan utgifne, D. 2*. Stockholm: J. P. Lindh, 1812.

Candréus, C. "Dealing with Mobility: Guilds and Tramping Journeymen in Seventeenth-Century Scandinavia." *Cultural and Social History* 21, no. 2 (2024): 163–184. doi:10.1080/14780038.2024.2304384.

Caracausi, A. "Textiles Manufacturing, Product Innovations and Transfers of Technology in Padua and Venice Between the Sixteenth and Eighteenth Centuries." In *Innovation and Creativity in Late Medieval and Early Modern European Cities*, edited by K. Davids and B. De Munck, 131–160. Farnham: Ashgate, 2014.

Christensen, D. C. *Det moderne projekt: Teknik & kultur i Danmark-Norge 1750-(1814)-1850*. Copenhagen: Gyldendal, 1996.

Clemente, A. "Innovation in the Capital City: Central Policies, Markets and Migrant Skills in Neapolitan Ceramic Manufacturing in the Eighteenth Century." In *Innovation and Creativity in Late Medieval and Early Modern European Cities*, edited by K. Davids and B. De Munck, 315–335. Farnham: Ashgate, 2014.

Clifford, H. "Concepts of Invention, Identity and Imitation in the London and Provincial Metal-Working Trades, 1750–1800." *Journal of Design History* 12, no. 3 (1999): 241–255. doi:10.1093/jdh/12.3.241.

Davids, K. "Apprenticeship and Guild Control in the Netherlands, c.1450–1800." In *Learning on the Shop Floor: Historical Perspectives on Apprenticeship*, edited by B. De Munck, S. L. Kaplan, and H. Soly, 65–84. New York and Oxford: Berghahn Books, 2007.

Davids, K., and B. De Munck. "Innovation and Creativity in Late Medieval and Early Modern European Cities: An Introduction." In *Innovation and Creativity in Late Medieval and Early Modern European Cities*, edited by K. Davids and B. De Munck, 1–33. Farnham: Ashgate, 2014.

De Munck, B. "Artisans as Knowledge Workers: Craft and Creativity in a Long Term Perspective." *Geoforum* 99 (2019): 227–237. doi:10.1016/j.geoforum.2018.05.025.

De Munck, B. "Shared Skills and Technologies of Community Formation: Artisanal Epistemologies, Secrecy and Governmentality in Long-term Context." In *The Republic of Skill: Artisan Mobility, Innovation, and the Circulation of Knowledge in Premodern Europe*, edited by D. Garrioch, 35–64. Leiden: Brill, 2022.

De Munck, B., and A. Romano. "Knowledge and the Early Modern City: An Introduction." In *Knowledge and the Early Modern City: A History of Entanglements*, edited by B. De Munck and A. Romano, 1–30. Abingdon, OX: Routledge, 2020.

De Munck, B., and H. Soly. "'Learning on the Shop Floor' in Historical Perspective." In *Learning on the Shop Floor: Historical Perspectives on Apprenticeship*, edited by B. De Munck, S. L. Kaplan, and H. Soly, 3–32. New York and Oxford: Berghahn Books, 2007.

De Vito, C. G., and A. Gerritsen. "Micro-Spatial Histories of Labour: Towards a New Global History." In *Micro-Spatial Histories of Global Labour*, edited by C. G. De Vito and A. Gerritsen, 1–28. Cham: Springer International, 2018.

Douhan, B. *Arbete, kapital och migration: Valloninvandringen till Sverige under 1600-talet*. Uppsala: Uppsala University, 1985.

Edgren, L. *Lärling-Gesäll-Mästare: Hantverk och hantverkare i Malmö 1750–1847*. Lund: Lund University, 1987.

Elers, J. *Stockholm. III. Delen*. Stockholm: H. Nordström Jr, 1801.

Epstein, S. R. "Labour Mobility, Journeyman Organisations and Markets in Skilled Labour in Europe, 14th-18th centuries." In *Le technicien dans la cité en Europe occidentale, 1250–1650*, edited by M. Arnoux and P. Monnet, 251–269. Rome: École française de Rome, 2004.

Evans, C., and G. Rydén. *Baltic Iron in the Atlantic World in the Eighteenth Century*. Leiden: Brill, 2007.

Evans, C., and A. Withey. "An Enlightenment in Steel? Innovation in the Steel Trades of Eighteenth-Century Britain." *Technology and Culture* 53, no. 3 (2012): 533–560. doi:10.1353/tech.2012.0116.

Felten, S., and C. von Oertzen. "Bureaucracy as Knowledge." *Journal for the History of Knowledge* 1, no. 1/Special Issue (2020): 1–16. doi:10.5334/jhk.18.

Florida, R. *The Rise of the Creative Class: And How it's Transforming Work, Leisure, Community and Everyday Life*. New York: Basic Books, 2002.

Fors, H. *The Limits of Matter: Chemistry, Mining and Enlightenment*. Chicago: The University of Chicago Press, 2015.

Fors, H., and J. Orrje. "Describing the World and Shaping the Self: Knowledge-Gathering, Mobility and Spatial Control at the Swedish Bureau of Mines." In *Transnational Cultures of Expertise: Circulating State-Related Knowledge in the 18th and 19th Centuries*, edited by L. Schilling and J. Vogel, 107–128. Berlin: De Gruyter Oldenbourg, 2019.

Gårdlund, T. "Teknik och tekniker i den tidiga svenska verkstadsindustrien." *Ekonomisk Tidskrift* 42, no. 4 (1940): 179–191. doi:10.2307/3438194.

Garrioch, D. "Introduction: Artisan Mobility and Innovation in Pre-Industrial Europe." In *The Republic of Skill: Artisan Mobility, Innovation, and the Circulation of Knowledge in Premodern Europe*, edited by D. Garrioch, 1–34. Leiden: Brill, 2022.

Gustafsson, S. "Rörlig kunskap och flyttbara färdigheter – skolad arbetskraft vid fästningsbygget Sveaborg på 1750-talet." *1700-tal: Nordic Journal for Eighteenth-Century Studies* 13 (2016): 31–53. doi:10.7557/4.3872.

Halverson, K. *Tools of the Trade: Medical Devices and Practice in Sweden and Denmark, 1855–1897*. Huddinge: Södertörns högskola, 2022.

Hammarström, I. *Stockholm i svensk ekonomi 1850–1914*. Stockholm: Stadsarkivet, 1970.

Harris, J. R. *Industrial Espionage and Technology Transfer: Britain and France in the Eighteenth Century*. Aldershot: Ashgate, 1998.

Hayen, M. *Stadens puls: En tidsgeografisk studie av hushåll och vardagsliv i Stockholm, 1760–1830*. Stockholm: Stads- och kommunhistoriska institutet, 2007.

Hayen, M., and K. Nyberg. "Konkursinstitutet i Stockholm 1720–1850." In *Ekonomisk kulturhistoria: Bildkonst, konsthantverk och scenkonst 1720–1850*, edited by K. Nyberg, 37–52. Stockholm: Kulturhistoriska Bokförlaget, 2017.

Heckscher, E. F. "De svenska manufakturerna under 1700-talet." *Ekonomisk Tidskrift* 39, no. 6 (1937): 153–221. doi:10.2307/3438478.

Heckscher, E. F. *Sveriges ekonomiska historia från Gustav Vasa D. 2 Det moderna Sveriges grundläggning, Halvbd 1*. Stockholm: Bonnier, 1949a.

Heckscher, E. F. *Sveriges ekonomiska historia från Gustav Vasa D. 2 Det moderna Sveriges grundläggning, Halvbd 2*. Stockholm: Bonnier, 1949b.

Hilaire-Pérez, L. "Invention and the State in 18th-Century France." *Technology and Culture* 32, no. 4 (1991): 911–931. doi:10.1353/tech.1991.0003.

Hilaire-Pérez, L. *L'invention technique au siècle des Lumières*. Paris: A. Michel, 2000.

Hilaire-Pérez, L. "Technology as a Public Culture in the Eighteenth Century: The Artisans' Legacy." *History of Science* 45, no. 2 (2007): 135–153. doi:10.1177/007327530704500202.

Hilaire-Pérez, L. "The Codification of Techniques: Between Bureaucracy and the Markets in Early Modern Europe from a Global Perspective." *Technology and Culture* 62, no. 2 (2021): 442–466. doi:10.1353/tech.2021.0056.

Hilaire-Pérez, L., and C. Lanoë. "Par-delà les pièges des discours, les techniques artisanales aux XVIIe et XVIIIe siècles." *Annales: Histoire, Sciences Sociales* 79, no. 3 (2024): 361–390. doi:10.1017/ahss.2024.60.

Hilaire-Pérez, L., and C. Rabier. "Self-Machinery? Steel Trusses and the Management of Ruptures in Eighteenth-Century Europe." *Technology and Culture* 54, no. 3 (2013): 460–502. doi:10.1353/tech.2013.0096.

Hilaire-Pérez, L., and C. Verna. "Dissemination of Technical Knowledge in the Middle Ages and the Early Modern Era: New Approaches and Methodological Issues." *Technology and Culture* 47, no. 3 (2006): 536–565. doi:10.1353/tech.2006.0179.

Hinners, L. *De fransöske handtwerkarne vid Stockholms slott 1693–1713: yrkesroller, organisation, arbetsprocesser*. Stockholm: Konstvetenskapliga institutionen, Stockholm University, 2012.

Hutchison, R. "Exploring an Early Cross-border Trade System: Norwegian Copper in the 18th Century." *Scandinavian Journal of History* 44, no. 3 (2019): 261–286. doi:10.1080/03468755.2018.1509805.

Isacson, M., and L. Magnusson. *Proto-industrialisation in Scandinavia: Craft Skills in the Industrial Revolution*. Leamington Spa: Berg, 1987.

Jakobsson, H. "Economic behaviour and social strategies in the Stockholm silk weaving industry, 1744–1831." In *Luxury, Fashion and the Early Modern Idea of Credit*, edited by K. Nyberg and H. Jakobsson, 97–119. Abingdon, OX: Routledge, 2021.

Jansson, M. *Making Metal Making: Circulation and Workshop Practices in the Swedish Metal Trades, 1730–1775*. Uppsala: Acta Universitatis Upsaliensis, 2017.

Jansson, M. "Global Materials, Contact Zones, and Imitation: The Making of Metal Commodities in Eighteenth-Century Sweden." In *Locating the Global: Spaces, Networks, and Interactions from the Seventeenth to the Twentieth Century*, edited by H. Weiss, 41–72. Berlin: De Gruyter, 2020.

Jansson, M. "Nipper och nyttigt smide i 1700-talets metallvaruhandel." In *Shopping i Stockholm: Sociala praktiker på gatunivå, 1700–1850*, edited by M. Hellsing and J. Ilmakunnas, 233–261. Stockholm: Stockholmia förlag, 2023.

Jansson, M., and G. Rydén. "Making and Trading Metals: A Narrative of Swedish Improvement." In *Political Reason and the Language of Change: Reform and Improvement in Early Modern Europe*, edited by A. Luna-Fabritius, E. Nokkala, M. Seppel, and K. Tribe, 218–238. Milton: Taylor & Francis Group, 2022.

Jansson, M., and G. Rydén. "Improving Swedish Steelmaking: Circulation and Localized Knowledge-Making in Early Modernity." *Technology and Culture* 64, no. 2 (2023): 515–542. doi:10.1353/tech.2023.0060.

Jansson, M., and G. Rydén. "Rörelser mot det industriella: Bergsbruk och historisk förändring." *Historisk Tidskrift* 144, no. 3 (2024): 360–388.

Jones, P. M. "In Search of the 'Invisible' Artisan during Europe's Industrial Enlightenment: James White, 1762–1825." *International Journal for the History of Engineering & Technology* 95, no. 2 (2025): 171–196. doi:10.1080/17581206.2025.2456275.

Jülich, S. "Johan Albrecht (Albert) Stille." In *Svenskt biografiskt lexikon*, Bd 33, edited by Å. Karlsson, 485–488. Stockholm: Svenskt biografiskt lexikon, 2007–2011.

Kjellander, R. "Thomas Lewis." In *Svenskt biografiskt lexikon*, Bd 22, edited by E. Grill and B. Lager-Kromnow, 668–670. Stockholm: Svenskt biografiskt lexikon, 1977–1979.

Kungl Vetenskapsakademien, *Kongl. Vetenskaps-Academiens handlingar för år 1838*. Stockholm: P.A. Norstedt & Söner, 1839.

Lagerquist, M. "Argent Haché – en illusion av gediget silver." *RIG – Kulturhistorisk tidskrift* 34, no. 1 (1951): 9–16.

Lagerquist, M. "Petter Ernst." In *Svenskt biografiskt lexikon*, Bd 14, edited by B. Hildebrand, 458–460. Stockholm: Samfundet för Svenskt biografiskt lexikons utgivande, 1953.

Lilja, S. *Tjuvehål och stolta städer: urbaniseringens kronologi och geografi i Sverige (med Finland) ca 1570-tal till 1810-tal*. Stockholm: Stads- och kommunhistoriska institutet, 2000.

Lindqvist, S. *Technology on Trial: The Introduction of Steam Power Technology into Sweden, 1715–1736*. Stockholm: Almqvist & Wiksell International, 1984.

Long, P. O. *Artisan/Practitioners and the Rise of the New Sciences, 1400–1600*. Corvallis, OR: Oregon State University Press, 2011.

Lucas, A. "Manufacturing Modernity: Innovations in Early Modern Europe—An Introduction." *Technology and Culture* 61, no. 4 (2020): 995–1016. doi:10.1353/tech.2020.0108.

Luu, L. B. *Immigrants and the Industries of London, 1500–1700*. Aldershot, Hants: Ashgate, 2005.
MacLeod, C. *Inventing the Industrial Revolution: The English Patent System, 1660–1800*. Cambridge: Cambridge University Press, 1988.
Melkersson, M. *Staten, ordningen och friheten: En studie av den styrande elitens syn på statens roll mellan stormaktstiden och 1800-talet*. Uppsala: Uppsala University, 1997.
Minard, P. "Colbertism Continued? The Inspectorate of Manufactures and Strategies of Exchange in Eighteenth-Century France." *French Historical Studies* 23, no. 3 (2000): 477–496. doi:10.1215/00161071-23-3-477.
Nyberg, K. *Köpes: ull, säljes: kläde: Yllemanufakturens företagsformer i 1780-talets Stockholm*. Uppsala: Uppsala University, 1992.
Nyberg, K. *Kommersiell kompetens och industrialisering: Norrköpings ylleindustriella tillväxt på Stockholms bekostnad 1780–1846*. Uppsala: Acta Universitatis Upsaliensis, 1999.
Nyberg, K., ed. *Till salu: Stockholms textila handel och manufaktur 1722–1846*. Stockholm: Stads- och kommunhistoriska institutet, 2010.
Nyberg, K. "The Production of International Fashion in State-Sponsored Manufactures in Sweden, 1740–1810. Part I." In *Fashion as an Economic Engine: Process and Product Innovation, Commercial Strategies, Consumer Behaviour*, edited by G. Nigro, 221–250. Florence: Firenze University Press, 2022.
Nyström, P. *Stadsindustriens arbetare före 1800-talet*. Stockholm: Tiden, 1955.
Ohlsson, B.-E. *Eskilstuna fristad: fristadsinrättningen i Eskilstuna före sammanslagningen med gamla staden 1771–1833*. Eskilstuna: Eskilstuna kommun, 1971.
Orrje, J. "Patriotic and Cosmopolitan Patchworks: Following a Swedish Astronomer into London's Communities of Maritime Longitude, 1759–60." In *Navigational Enterprises in Europe and Its Empires, 1730–1850*, edited by R. Dunn and R. Higgitt, 89–110. Basingstoke: Palgrave Macmillan, 2016.
Orrje, J. "The Logistics of the Republic of Letters: Mercantile Undercurrents of Early Modern Scholarly Knowledge Circulation." *British Journal for the History of Science* 53, no. 3 (2020): 351–369. doi:10.1017/S0007087420000242.
Osborn, W. *Essays on the Practice of Midwifery, in Natural and Difficult Labours*. London: Cadell/ Johnson, 1792.
Pérez, L. "Inventing in a World of Guilds: Silk Fabrics in Eighteenth-Century Lyon." In *Guilds, Innovation and the European Economy, 1400–1800*, edited by S. R. Epstein and M. Prak, 232–263. Cambridge: Cambridge University Press, 2008.
Pérez, L. "Technology, Curiosity and Utility in France and in England in the Eighteenth Century." In *Science and Spectacle in the European Enlightenment*, edited by B. Bensaude-Vincent and C. Blondel, 25–42. Aldershot: Ashgate, 2008.
Petterson, J.-E. "Samuel Owen." In *Svenskt biografiskt lexikon*, Bd 28, edited by G. Nilzén, 459–462. Stockholm: Svenskt biografiskt lexikon, 1992–1994.
Prak, M., C. Haru Crowston, B. De Munck, C. Kissane, C. Minns, R. Schalk, and P. Wallis. "Access to the Trade: Monopoly and Mobility in European Craft Guilds in the Seventeenth and Eighteenth Centuries." *Journal of Social History* 54, no. 2 (2020): 421–452. doi:10.1093/jsh/shz070.
Puetz, A. "Design Instruction for Artisans in Eighteenth-Century Britain." *Journal of Design History* 12, no. 3 (1999): 217–239. doi:10.1093/jdh/12.3.217.
Raj, K. *Relocating Modern Science: Circulation and the Construction of Knowledge in South Asia and Europe, 1650–1900*. Houndmills, Basingstoke, Hampshire: Palgrave Macmillan, 2007.
Ranestad, K. "Copper Trade and Production of Copper, Brass and Bronze Goods in the Oldenburg Monarchy: Copperworks and Copper Users in the Eighteenth Century." *Scandinavian Economic History Review* 67, no. 2 (2019): 190–209. doi:10.1080/03585522.2019.1566767.
Reith, R. "Circulation of Skilled Labour in Late Medieval and Early Modern Central Europe." In *Guilds, Innovation and the European Economy, 1400–1800*, edited by S. R. Epstein and M. Prak, 114–142. Cambridge: Cambridge University Press, 2008.
Riello, G. "Strategies and Boundaries: Subcontracting and the London Trades in the Long Eighteenth Century." *Enterprise & Society* 9, no. 2 (2008): 243–280. doi:10.1093/es/khn004.

Roberts, L. "The Circulation of Knowledge in Early Modern Europe: Embodiment, Mobility, Learning and Knowing." In *History of Technology*, edited by I. Inkster, 31 vols. 47–68. London: Bloomsbury, 2012.
Roberts, L., and S. Schaffer. "Preface." In *The Mindful Hand: Inquiry and Invention from the Late Renaissance to Early Industrialisation*, edited by L. Roberts, S. Schaffer, and P. Dear, XIII–XXVII. Amsterdam: Koninkliijke Nederlandse Akademie van Wetenschappen, 2007.
Roche, D. *A History of Everyday Things: The Birth of Consumption in France, 1600–1800*. Cambridge: Cambridge University Press, 2000.
Ronnestam, B. J. *Fredmans tid: frihetstiden – guldåldern för Stockholms urmakare*. Stockholm: Carlsson, 2013.
Rydberg, S. *Svenska studieresor till England under Frihetstiden*. Uppsala: Uppsala University, 1951.
Rydén, G. "Eskilstuna *Fristad*: The Beginnings of an Urban Experiment." In *Sweden in the Eighteenth-Century World: Provincial Cosmopolitans*, edited by G. Rydén, 123–144. Farnham: Ashgate, 2013a.
Rydén, G. "The Enlightenment in Practice: Swedish Travellers and Knowledge About the Metal Trades." *1700-tal: Nordic Journal for Eighteenth-Century Studies* 10 (2013b): 63–86. doi:10.7557/4.2621.
Salvius, L. *Beskrifning öfver Sveriget, Första Tomen om Upland, af Lars Salvius*. Stockholm: Salvius: Horn: Röpke, 1741.
Schaffer, S., L. Roberts, K. Raj, and J. Delbourgo. "Introduction." In *The Brokered World: Go-Betweens and Global Intelligence, 1770–1820*, edited by S. Schaffer, L. Roberts, K. Raj, and J. Delbourgo, ix–xxxviii. Sagamore Beach, Mass: Science History Publications, 2009.
Schröderstierna, S. *Berättelser om de finare järn- stål- och metallfabrikerna i Sverige åren 1754–1759, I, II*. Edited by Gösta Malmborg. Stockholm: Generalstabens litografiska anstalt, 1925.
Simonsson, I. "Daniel Asplund." In *Svenskt biografiskt lexikon*, Bd 2, edited by B. Boëthius, 402–404. Stockholm: Bonnier, 1920.
Smith, P. H. *The Body of the Artisan: Art and Experience in the Scientific Revolution*. Chicago: University of Chicago Press, 2004.
Smith, P. H. "Making as Knowing: Craft as Natural Philosophy." In *Ways of Making and Knowing: The Material Culture of Empirical Knowledge*, edited by P. H. Smith, A. R. W. Meyers, and H. J. Cook, 17–47. Ann Arbor: University of Michigan Press, 2014.
Smith, P. H. "Nodes of Convergence, Material Complexes, and Entangled Itineraries." In *Entangled Itineraries: Materials, Practices, and Knowledges Across Eurasia*, edited by P. H. Smith, 5–24. Pittsburgh: University of Pittsburgh Press, 2019.
Söderberg, J., U. Jonsson, and C. Persson. *A Stagnating Metropolis: The Economy and Demography of Stockholm, 1750–1850*. Cambridge: Cambridge University Press, 1991.
Söderlund, E. *Stockholms hantverkarklass 1720–1772: sociala och ekonomiska förhållanden*. Stockholm: Norstedts, 1943.
Sonenscher, M. *Work and Wages: Natural Law, Politics and the Eighteenth-Century French Trades*. 1989. Reiss. ed. Cambridge: Cambridge University Press, 2012.
Styles, J. "Product Innovation in Early Modern London." *Past & Present* 168, no. 1 (2000): 124–169. doi:10.1093/past/168.1.124.
Torstendahl, R. *Teknologins nytta: Motiveringar för det svenska tekniska utbildningsväsendets framväxt framförda av riksdagsmän och utbildningsadministratörer 1810–1870*. Uppsala: Uppsala University, 1975.
Valleriani, M. "The Epistemology of Practical Knowledge." In *The Structures of Practical Knowledge*, edited by M. Valleriani, 1–19. Cham: Springer International Publishing, 2017.
Wallis, P. "Apprenticeship and Training in Premodern England." *The Journal of Economic History* 68, no. 3 (2008): 832–861. doi:10.1017/S002205070800065X.
Wallis, P. "Between Apprenticeship and Skill: Acquiring Knowledge Outside the Academy in Early Modern England." *Science in Context* 32, no. 2 (2019): 155–170. doi:10.1017/S0269889719000164.
Wallis, P., and C. Wright. "Evidence, Artisan Experience, and Authority in Early Modern England." In *Ways of Making and Knowing: The Material Culture of Empirical Knowledge*,

edited by P. H. Smith, A. R. W. Meyers, and H. J. Cook, 138–163. Ann Arbor: University of Michigan Press, 2014.
Westrin, T., ed. *Nordisk familjebok: Konversationslexikon och Realencyklopedi*, vol. 18. Stockholm: Nordisk familjeboks förlags aktiebolag, 1913.
Wikström, L. "Peter Momma." In *Svenskt biografiskt lexikon*, Bd 25, edited by G. Nilzén, 667–669. Stockholm: Svenskt biografiskt lexikon, 1985–1987.
Wollin, N. G. *Svenska stilgjuteriet: Peter Mommas skapelse*. Stockholm: Gebers, 1947.